

CHAPTER 7

EXPECTATION

7.1 INTRODUCTION

The probabilistic concepts discussed in Chapters 5 and 6 could have also been developed without using random variables but in a clumsy and awkward way. Random variables were used there as a convenient tool of describing large classes of events. Indeed, once we considered events of the form $\{a < X \leq b\}$, it was quite natural to reduce the analysis to even simpler events $\{X \leq t\}$. Then the probability of such an event, regarded as a function of the argument t [i.e., the cumulative distribution function (cdf) of X] turned out to carry all information about the distribution of X . And once the notion of the cdf was introduced, it was natural to look for classes of cdf's that allow for simple description (hence, the definition of discrete and continuous random variables).

In this chapter, we introduce the notion of *expectation*. Expectation, or the *expected value* of a random variable, cannot be formulated without the concept of random variables.

The intuitive content of the notion of expectation is as follows: Consider a random variable X , defined on some sample space $\mathcal{S}$. An experiment consists of a random selection of a point s of $\mathcal{S}$, and $X(s)$ is interpreted as the gain (loss, if $X(s) < 0$) of a hypothetical gambler.

If the experiment is repeated n times, the sample space becomes $\mathcal{S}^n = \mathcal{S} \times \cdots \times \mathcal{S}$ (n times). An outcome is $s^{(n)} = (s_1, \dots, s_n)$, with $s_i \in \mathcal{S}, i = 1, \dots, n$. The gambler's accumulated gain becomes $X(s_1) + X(s_2) + \cdots + X(s_n)$, traditionally written as $X_1 + X_2 + \cdots + X_n$, with $X_i = X(s_i)$ being the outcome of the i th trial.

The average gain per gamble now becomes $(X_1 + \cdots + X_n)/n$. As n becomes larger, this average fluctuates less and less, tending to stabilize at some value, which we call the expectation of X and denote by $E(X)$.

It is not difficult to calculate this value in case of a discrete random variable. Indeed, if X assumes values $x_1, x_2, \dots, x_m$ with probabilities $p_i = P\{X = x_i\}, i = 1, 2, \dots, m$ and if the

experiment is to be repeated n times, then $\{X = x_i\}$ will occur n_i times ($i = 1, \dots, m$), and $n_1 + n_2 + \dots + n_m = n$. According to the frequency interpretation of probability, the ratio n_i/n tends to $p_i = P\{X = x_i\}$ as the number n of repetitions of the experiments increases. The total gain $X_1 + \dots + X_n$ equals $x_1 n_1 + x_2 n_2 + \dots + x_m n_m$, and the average gain per experiment,

$$x_1 \frac{n_1}{n} + x_2 \frac{n_2}{n} + \dots + x_m \frac{n_m}{n},$$

tends to

$$x_1 p_1 + x_2 p_2 + \dots + x_m p_m.$$

We will use the last quantity, that is,

$$E(X) = x_1 P\{X = x_1\} + \dots + x_m P\{X = x_m\} \quad (7.1)$$

as a formal definition of an expected value $E(X)$ of a discrete random variable X . The definition will later be extended to also cover random variables that are not discrete.

7.2 EXPECTED VALUE

For simplicity, the definition of an expected value of a random variable X is now formulated separately for the discrete and continuous random variables.

Definition 7.2.1 If X is a discrete random variable that assumes values $x_1, x_2, \dots$ with probabilities $P\{X = x_i\}$, $i = 1, 2, \dots$, then the expected value of X is defined as

$$E(X) = \sum_i x_i P\{X = x_i\}, \quad (7.2)$$

provided that

$$\sum_i |x_i| P\{X = x_i\} < \infty. \quad (7.3)$$

If X is a continuous random variable with the density $f(x)$, then the *expected value* of X is defined as

$$E(X) = \int_{-\infty}^{+\infty} x f(x) dx, \quad (7.4)$$

provided that

$$\int_{-\infty}^{+\infty} |x| f(x) dx < \infty. \quad (7.5)$$

□

Definition 7.2.1 covers formula (7.1) as a special case, since if X assumes only finitely many values, condition (7.3) is always satisfied. Conditions (7.3) and (7.5) are necessary to eliminate the situation where the value of an infinite sum depends on the order of summation (or integration). The absolute convergence (or absolute integrability) guarantees that the expected value is unambiguously defined.

The other symbols used for the expected value $E(X)$ in probabilistic and statistical literature may be EX , μ_X , m_X , or simply μ , m , if it is clear which random variable is being studied. In physics, the symbol used is $\langle X \rangle$. Regarding terminology, the expected value is also called the expectation, or the mean. Moreover, the expected value is often associated not with X , but with its distribution. Consequently, one sometimes uses symbols such as μ_F or m_F , where F is a cdf.

We begin with a series of examples of expectations for discrete random variables.

■ EXAMPLE 7.1

Let A be some event, and let $p = P(A)$. Imagine that you are to receive \$1 if A occurs and nothing if A does not occur (i.e., if A^c occurs). Let X be the random variable defined as “the amount you receive.” In this case, X assumes only two values, and its distribution is given by the array

Value	0	1
Probability	$1 - p$	p

According to formula (7.2),

$$E(X) = 0 \times (1 - p) + 1 \times p = p.$$

The result is quite intuitive: if you play such a game over and over, then your average outcome per game will equal the probability of winning, p .

Observe that X can also be described as “the number of occurrences of A in a single experiment.” We will use this interpretation later.

■ EXAMPLE 7.2

Suppose now that you are to win \$1 if A occurs, but lose \$1 (i.e., “win” -1) if A does not occur. Then the distribution of Y , the random variable describing your winnings, is

Value	-1	1
Probability	$1 - p$	p

and $E(Y) = -1 \times (1 - p) + 1 \times p = 2p - 1$. Thus, $E(Y) > 0$ if and only if $p > 0.5$, which agrees with our intuition: the game is favorable if the probability p of winning exceeds 0.5.

■ EXAMPLE 7.3 Expected Value of Binomial Distribution

The binomial random variable was defined in Example 5.7 as the total number of successes in n independent Bernoulli trials, where the probability of a success in any single trial is p . The possible values of the binomial random variable are $0, 1, \dots, n$ with probabilities given by (5.13). Consequently,

$$\begin{aligned}
 E(X) &= \sum_{k=0}^n k \binom{n}{k} p^k (1-p)^{n-k} = \sum_{k=1}^n k \frac{n!}{k!(n-k)!} p^k (1-p)^{n-k} \\
 &= np \sum_{k=1}^n \frac{(n-1)!}{(k-1)!(n-1-(k-1))!} p^{k-1} (1-p)^{n-1-(k-1)} \\
 &= np \sum_{j=0}^{n-1} \binom{n-1}{j} p^j (1-p)^{n-1-j} = np,
 \end{aligned}$$

where in the last sum we introduced the new index of summation $j = k - 1$. The last sum turns out to be a Newtonian expansion of $[p + (1 - p)]^{n-1} = 1$.

■ EXAMPLE 7.4 Expectation of Geometric Distribution

Let an event A (“success”) have probability p . We keep repeating independently the experiments until A occurs for the first time, with X being the number of trials up to and including the first success. As already mentioned (see Example 5.8), X is a random variable with geometric distribution and $P\{X = k\} = (1 - p)^{k-1}p$ for $k = 1, 2, \dots$. Find $E(X)$.

SOLUTION. Letting $q = 1 - p$, we write

$$\begin{aligned} E(X) &= \sum_{k=1}^{\infty} kP\{X = k\} = \sum_{k=1}^{\infty} kq^{k-1}p = p \sum_{k=1}^{\infty} \frac{d}{dq}(q^k) \\ &= p \frac{d}{dq} \sum_{k=1}^{\infty} q^k = p \frac{d}{dq} \left(\frac{q}{1-q} \right) = p \times \frac{1}{(1-q)^2} = \frac{1}{p}. \end{aligned}$$

The answer is intuitive. If, for example, the probability of success is $p = 0.2$, then on average, every fifth trial is a success, and Therefore, the average number of trials until the next success is 5, which agrees with formula (7.6).

The following theorem provides a very useful interpretation of the expected value of a random variable:

Theorem 7.2.1 *Let X be a random variable that assumes only nonnegative integer values. Then*

$$E(X) = \sum_{n=0}^{\infty} P\{X > n\}. \quad (7.6)$$

Proof: For $p_n = P\{X = n\}$ we have

$$\begin{aligned} E(X) &= p_1 + 2p_2 + 3p_3 + \dots \\ &= p_1 + p_2 + p_3 + \dots \\ &\quad + p_2 + p_3 + \dots \\ &\quad + p_3 + \dots \\ &= P\{X > 0\} + P\{X > 1\} + P\{X > 2\} + \dots, \end{aligned}$$

where a change of order of summation is allowed, since all terms are nonnegative. $\square$

■ EXAMPLE 7.5

Let us use formula (7.6) to find the expectation of the geometric distribution from Example 7.4. We have

$$P\{X > n\} = P\{\text{only failures in the first } n \text{ trials}\} = q^n,$$

or, if this reasoning appears suspiciously simple and one puts more trust in algebra,

$$P\{X > n\} = \sum_{k=n+1}^{\infty} P\{X = k\} = \sum_{k=n+1}^{\infty} q^{k-1}p = p \frac{q^n}{1-q} = q^n.$$

Consequently,

$$E(X) = \sum_{n=0}^{\infty} q^n = \frac{1}{1-q} = \frac{1}{p}.$$

The expected value in Example 7.5 was finite. Such situations are most common in practice, as otherwise, the very concept of expected value would make no sense. However, there are cases of both practical and theoretical significance, when the condition (7.3) or (7.5) is not met.

To be more precise, one should distinguish between two cases. We will illustrate the possibilities of condition (7.3) involving the series; the situation with an integral is analogous. Let us consider separately sums $\sum x_i P\{X = x_i\}$ involving positive and negative terms, that is, $U^+ = \sum [\max(x_i, 0)] \times P\{X = x_i\}$ and $U^- = -\sum [\min(x_i, 0)] \times P\{X = x_i\}$. It is clear that if the series (7.3) diverges, then U^+ , U^- , or both must be infinite, for otherwise, the series of absolute values (7.3), equal to $U^+ + U^-$, would be finite. If only U^+ is infinite, we can say that $E(X) = +\infty$; similarly, if only U^- is infinite, we can say that $E(X) = -\infty$. It is only when both U^+ and U^- are infinite that we have the expression of the form $\infty - \infty$, to which we cannot assign any value. In the last case, we say that $E(X)$ *does not exist*.

In the following example, random variable will have infinite expectation:

■ EXAMPLE 7.6 Petersburg Paradox

Suppose that you have the possibility of participating in the following game. A fair coin is tossed repeatedly until a head occurs. If the first head occurs at the k th toss, you win 2^k dollars. Clearly, this is a game in which you cannot lose, so the question is: What is the amount you should be willing to pay to participate in such game?

SOLUTION. The most common argument offered in such cases is that one should be willing to pay the fee as long as the expected winnings exceed or equal the fee for participation in the game. Then, “in the long run,” one should come even or ahead.

Let X denote the winnings in the game (not counting the fee for participation in the game). Then the event $\{X = 2^k\}$ occurs if the first $k - 1$ tosses are tails, and the k th toss is heads; chances of this event are $1/2^k$. Consequently, we have

$$E(X) = 2 \times \frac{1}{2} + 2^2 \times \frac{1}{2^2} + 2^3 \times \frac{1}{2^3} + \cdots = \infty.$$

Because the expected winnings in this game are infinite, one should be willing to pay an infinite fee for the right to play. However, in any particular game, the winnings will always be finite, so one is certain to lose.

This phenomenon, discovered in eighteenth century, was named the Petersburg paradox. Attempts to solve it led to the introduction of the concept of utility. Without going into detail, it is postulated that the utility of money—the value a person attaches to a given amount of money—is not constant but depends on how much money the person already has. This appears intuitively acceptable: \$20 may seem to be a lot for someone who has nothing but does not seem like very much to a millionaire.

If $u(x)$ is the utility of \$ x , then the above postulate implies that $u(x)$ is not proportional to x . The *expected utility* from participating in the game is

$$E[u(X)] = \sum_{k=1}^{\infty} u(2^k) \times \frac{1}{2^k},$$

a quantity that may be finite for some “plausible” choices of utility function $u(x)$. In particular, Bernoulli suggested $u(x) = c \log x$.

It may be worthwhile to give the following “mechanical” interpretation of $E(X)$. Consider a discrete random variable, with possible values x_i and the corresponding probabilities $p_i = P\{X = x_i\}, i = 1, 2, \dots$

Let us construct a mechanical device corresponding to the distribution of X . Consider an infinitely thin and rigid wire extended from $-\infty$ to ∞ (this is an abstraction that physicists use: an ideal void, a perfectly black body, etc.). At points with coordinates $x_1, x_2, \dots$ we attach weights that are proportional to probabilities p_i (see Figure 7.1 where the sizes of the dots symbolize the variability in weight). Then $E(X)$ is the coordinate of the point at which the whole device would balance, or its center of gravity.

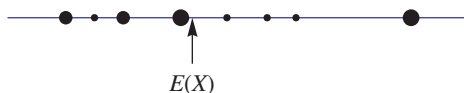


Figure 7.1 Interpretation of expected value of a discrete random variable.

A similar interpretation also holds for continuous random variables. In this case, we have to imagine that an infinitely rigid metal sheet of uniform weight is cut into the shape of density $f(x)$. Then, $E(X)$ is the coordinate of the point at which the figure will balance if laid flat (see Figure 7.2).

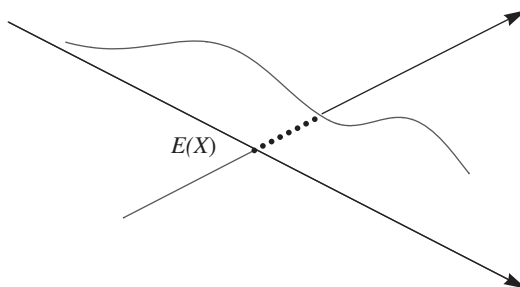


Figure 7.2 Interpretation of expected value of a continuous random variable.

We will now give some examples of expectations of continuous type random variables.

■ EXAMPLE 7.7

Let X have a distribution uniform on interval $[A, B]$ so that (see Example 5.14)

$$f(x) = \begin{cases} \frac{1}{B-A} & \text{for } A \leq x \leq B \\ 0 & \text{otherwise.} \end{cases}$$

According to the interpretation given above, the metal sheet figure is just a rectangle. Hence, it will balance if supported in the middle between A and B . Thus, we can expect that $E(X) = (A + B)/2$. Indeed, the computations give

$$E(X) = \int_{-\infty}^{+\infty} x f(x) dx = \int_A^B \frac{x}{B-A} dx = \frac{B^2 - A^2}{2(B-A)} = \frac{A+B}{2}.$$

EXAMPLE 7.8

We will now find the expected value of random variable X with exponential distribution with density $f(x) = \lambda e^{-\lambda x}$ for $x \geq 0$ and 0 otherwise.

SOLUTION. Integrating by parts, we obtain

$$E(X) = \int_0^{\infty} x \lambda e^{-\lambda x} dx = -\frac{1}{\lambda} e^{-\lambda x} \Big|_0^{\infty} = \frac{1}{\lambda}.$$

Thus, if the lifetime (in hours) of a certain type of electric bulb has a density $0.001e^{-0.001t}$ for $t > 0$, then the expected lifetime equals $(0.001)^{-1} = 1,000$ hours.

EXAMPLE 7.9

Find the expectation of a normal random variable with the density

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}.$$

SOLUTION. We substitute $z = (x - \mu)/\sigma$, $dx = \sigma dz$, obtaining

$$\begin{aligned} E(X) &= \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{+\infty} x e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} (\mu + z\sigma) e^{-\frac{z^2}{2}} dz \\ &= \mu \times \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} e^{-\frac{z^2}{2}} dz + \frac{\sigma}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} z e^{-\frac{z^2}{2}} dz = \mu. \end{aligned}$$

The first integral in the last line equals μ (since the standard normal density integrates to 1), while the second integral is 0 because the integrand is an odd function. So the integrals over the positive and negative parts of the z -axis (each being finite) cancel each other. Thus, μ is the expected value of the random variable with distribution $N(\mu, \sigma^2)$.

We will next derive an analogue of formula (7.6) for an expected value of a nonnegative continuous random variable.

Theorem 7.2.2 *Let T be a nonnegative random variable of continuous type, with density $f(t)$ and cdf $F(t)$, so that $f(t) = F(t) = 0$ for $t < 0$. Then*

$$E(T) = \int_0^{\infty} [1 - F(t)] dt.$$

Proof: Replacing t with $\int_0^t dx$ and changing the order of integration, we have

$$\begin{aligned} E(T) &= \int_0^{\infty} x f(x) dx = \int_0^{\infty} \left(\int_0^x dt \right) f(x) dx = \int_0^{\infty} \int_t^{\infty} f(x) dx dt \\ &= \int_0^{\infty} [1 - F(t)] dt. \end{aligned}$$

□

■ **EXAMPLE 7.10**

Let us compute again the expectation of a random variable with $\text{EXP}(\lambda)$ distribution. We have here $F(x) = 1 - e^{-\lambda x}$ for $x > 0$. Using Theorem 7.2.2, we obtain $E(X) = \int_0^\infty e^{-\lambda x} dx = 1/\lambda$.

■ **EXAMPLE 7.11**

Finally, let us consider the Cauchy distribution with the density given by

$$f(x) = \frac{1}{\pi} \times \frac{1}{1+x^2}.$$

It is easy to check that the function $f(x)$ is a density of a random variable:

$$\frac{1}{\pi} \int_{-\infty}^{+\infty} \frac{dx}{1+x^2} = \frac{1}{\pi} \arctan x \Big|_{-\infty}^{+\infty} = 1.$$

Regarding expectation, the positive part of the defining integral equals

$$\frac{1}{\pi} \int_0^{+\infty} \frac{x dx}{1+x^2} = \frac{1}{\pi} \times \frac{1}{2} \log(1+x^2) \Big|_0^{+\infty} = +\infty.$$

In a similar fashion, we check that $\int_{-\infty}^0 x f(x) dx = -\infty$. Hence, the expected value of Cauchy distribution does not exist.

PROBLEMS

- 7.2.1** Variable X assumes values 0, 1, 2, and 3 with probabilities 0.3, a , 0.1, and b , respectively. Find a and b if: **(i)** $E(X) = 1.5$. **(ii)** $E(X) = m$. First determine all possible values of m .
- 7.2.2** The density of the lifetime T of some part of electronic equipment is $f(t) = \lambda^2 t e^{-\lambda t}$, $t > 0$. Find $E(T)$.
- 7.2.3** Suppose there are $k = 10$ types of toys (plastic animals, etc.) to be found in boxes of some cereal. Assume that there is a toy in every box and that each type of toy occurs with equal frequency. **(i)** Find $E(X)$, where X is the number of boxes you must buy until you collect three different types of toys. **(ii)** Suppose that your little sister is collecting the cereal toys and would be happy to get the whole set of 10 different types. Find $E(X)$, where X is the number of boxes you must buy until she gets a complete collection.
- 7.2.4** Find the expected value $E(|X|)$, where X is a normal random variable with parameters $\mu = 0$ and σ^2 (the distribution of $|X|$ is called *folded normal* or *half normal*).
- 7.2.5** Let $X_1, X_2, \dots, X_n$ be independent, each with the same distribution. Find $E[\min(X_1, \dots, X_n)]$ if the distribution of variables is **(i)** $\text{EXP}(\lambda)$. **(ii)** $U[a, b]$. [Hint: First find the cdf (or density) of the random variable whose expectation is being computed.]
- 7.2.6** Let X be a nonnegative continuous random variable with hazard rate $h(t) = t$. Find $E(X)$.
- 7.2.7** Random variables X and Y are jointly distributed with density $f(x, y) = 12(x - y)^2$ for $0 \leq x < y \leq 1$ and 0 otherwise. Find $E(X)$ and $E(Y)$.

- 7.2.8** A point is randomly selected from the interval $(0, 1)$. Find the expected length of the smaller part of the interval.
- 7.2.9** Show that if the expectation of a continuous type random variable X exists and the density $f(x)$ of X satisfies the condition $f(x) = f(2a - x)$ for all $x \geq 0$, then $E(X) = a$.

7.3 EXPECTATION AS AN INTEGRAL*

Definition 7.2.1 of expected value in the discrete and continuous cases covers most situations occurring in practice. Many textbooks do not provide the general definition, and some do not even mention the fact that a general definition exists.

One of the consequences of such an omission is that one could come to the conclusion that the symbol $E(X + Y)$ makes no sense if X is discrete and Y is continuous (since $X + Y$ is neither a continuous nor discrete random variable). In fact, the expectation of a random variable is generally defined as an integral of a random variable, treated as a function on the sample space $\mathcal{S}$. We will now sketch this definition.

Riemann Integral

Let us start by briefly recalling the definition of an “ordinary” (i.e., Riemann) integral $\int_a^b g(x) dx$. For the moment assume that the function g is continuous on $[a, b]$. We first choose a sequence of partitions of the interval $[a, b]$. To simplify the presentation, suppose that the n th partition divides the interval $[a, b]$ into 2^n equal parts so that the k th point of the n th partition is

$$x_{n,k} = a + \frac{k}{2^n}(b - a), \quad k = 0, 1, \dots, 2^n.$$

Let $g_{n,k}^{(-)}$ be the minimum of function g in the k th interval, that is,

$$g_{n,k}^{(-)} = \min_{x_{n,k-1} \leq x \leq x_{n,k}} g(x).$$

We now form the (lower) sum approximating the integral

$$\underline{S}_n = \sum_{k=1}^{2^n} g_{n,k}^{(-)}(x_{n,k} - x_{n,k-1}) = \frac{b-a}{2^n} \sum_{k=1}^{2^n} g_{n,k}^{(-)}. \quad (7.7)$$

The upper sum $\overline{S}_n$ is defined in the same way, with the minimum in (7.7) replaced by the maximum, say $g_{n,k}^{(+)}$.

It is not difficult to show that $\underline{S}_n \leq \overline{S}_n$ for all n , and that the sequences $\{\underline{S}_n\}$ and $\{\overline{S}_n\}$ are monotone, the first increasing and the second decreasing. If they converge to the same limit, say S , then this limit is called the Riemann integral of g , and it is denoted by $\int_a^b g(x) dx$.

This is the essence of the definition; the details may vary in two respects. First, instead of dividing the interval $[a, b]$ into 2^n equal parts in the n th partition, we can take any points

$$a = x_{n,0} < x_{n,1} < \dots < x_{n,k_n} = b$$

as long as

$$\lim_{n \rightarrow \infty} \max_{1 \leq i \leq k_n} (x_{n,i} - x_{n,i-1}) = 0$$

(this implies that we must have $k_n \rightarrow \infty$).

Second, instead of taking the minimum and the maximum of g over the subintervals of partitions (which need not exist if g is not continuous), we can take the value of function g at a selected point in each subinterval. We then have one partial sum (instead of lower and upper sums), and the requirement is that the limits of these partial sums exist and are the same regardless of the choice of the partitions and of the intermediate points.

The main results of the theory built on this definition are very well known: continuous functions are integrable on closed finite intervals and if g has antiderivative G (i.e., $G' = g$), then

$$\int_a^b g(x) dx = G(b) - G(a).$$

The Riemann integral over an infinite range, $\int_{-\infty}^{+\infty} g(x) dx$, is defined in the usual manner through a limiting passage; we omit the details, which can be found in most calculus texts.

We recall the definition of the Riemann integral in order to better stress the differences and also the analogy between the principles of definition of the Lebesgue and Riemann integrals. The difference between the two definitions seems small and, at first, not essential. Yet the concept of the Lebesgue integral is of tremendous consequence, allowing us at once to free the concept of integral of all inessential constraints, and to stress its most crucial and significant features.

Lebesgue Integral

Again, assume at first that the function g , defined on the interval $[a, b]$, is continuous, and let $A < g(x) < B$. Instead of partitioning the interval $[a, b]$ on the x -axis, let us partition the interval $[A, B]$ on the y -axis. For simplicity, let us again take the partitions into 2^n equal parts,

$$A = y_{n,0} < y_{n,1} < \cdots < y_{n,2^n} = B,$$

where

$$y_{n,k} = A + \frac{k}{2^n}(B - A), \quad k = 0, 1, \dots, 2^n.$$

The lower sum approximating the integral can now be written as

$$\underline{S}_n^* = \sum_{k=0}^{2^n-1} y_{n,k} \times l(C_{n,k}), \quad (7.8)$$

where l stands for length and $C_{n,k}$ is the set of points x where the function g lies between $y_{n,k}$ and $y_{n,k+1}$; more precisely,

$$C_{n,k} = \{x : y_{n,k} \leq g(x) < y_{n,k+1}\}, \quad k = 0, 1, \dots, 2^n - 1.$$

The upper sum $\overline{S}_n^*$ is defined in a similar way, with $y_{n,k}$ in (7.8) replaced by $y_{n,k+1}$.

The lower sums $\underline{S}_n$ and $\underline{S}_n^*$ for Riemann and Lebesgue integrals of the same functions are illustrated as parts (R) and (L) in Figure 7.3.

The Lebesgue integral is now defined in the same way as the Riemann integral, namely as the common limit (if it exists) of the upper and lower sums. Again, we omit the details of the extension of the definition to an infinite range.

We may capture the difference in two definitions as follows: In the Riemann integral, when we partition the x -axis, we control the intervals on the x -axis, while the maxima and minima over these intervals depend on function g . In the Lebesgue integral, when we partition the y -axis, we control the values, but corresponding sets on the x -axis are determined by the function g . The distinction was summarized by Lebesgue, as a comparison of two

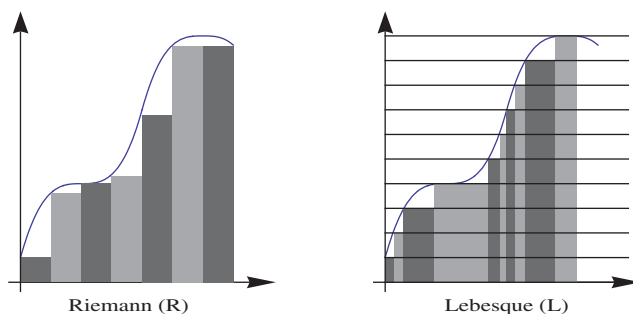


Figure 7.3 Approximating sums for Riemann and Lebesgue integrals.

cashiers who, at the end of the day, have to count the cash in their drawers. The “Riemann” cashier takes the money out in the order it came in. She counts “A five-dollar bill, and a penny makes 5.01; then I have a quarter, which makes 5.26; then . . .,” and so on. On the other hand, the “Lebesgue” cashier proceeds systematically: “I have n_1 twenty-dollar bills, which makes $20n_1$ dollars. Then I have n_2 ten-dollar bills, which together makes $20n_1 + 10n_2$ dollars . . .,” and so on.

One of the main results in the theory of the Lebesgue integral is that if a function is Riemann integrable, then it is also Lebesgue integrable, and both integrals are equal. Thus, we may use standard calculus to compute Lebesgue integrals, whenever it is applicable. However, it is important to point out that the class of functions that are Lebesgue integrable is substantially wider than the class of functions that are Riemann integrable. To see that it is so, consider the following example:

■ EXAMPLE 7.12

Let g be a function continuous on a closed finite interval $[a, b]$ so that g is Riemann integrable. It is clear that if we modify g at a single point x_0 by defining

$$g^*(x) = \begin{cases} g(x) & \text{if } x \neq x_0 \\ c & \text{if } x = x_0, \end{cases}$$

where c is any number such that $c \neq g(x_0)$, then the function $g^*(x)$ is Riemann integrable and $\int_a^b g(x) dx = \int_a^b g^*(x) dx$. We can modify in this way the function g at any *finite* set of points, and we will still have a Riemann integrable function, with the same integral as the original function g . However, if we modify g in such a way at a *countable* set of points, then the resulting function may not be Riemann integrable. This is because for a modification at finitely many points, say N , the sums approximating an integral of g and an integral of g^* differ in, at most, N terms. The difference tends to zero, since the sum of lengths of intervals on which the functions g and g^* differ tends to zero. But this is not necessarily the case where g and g^* differ at countably many points. Thus, g^* may be not integrable in the Riemann sense.

However, g^* is integrable in Lebesgue sense because two sets that differ by a countable set of points have the same length. So the sums approximating the Lebesgue integral are the same for g and g^* .

We will now present the two generalizations of the concept of integral introduced thus far. The first generalization concerns both the Riemann and Lebesgue integrals, leading in a natural way to the corresponding concepts of Riemann–Stieltjes and Lebesgue–Stieltjes

integrals. The second generalization, more important for our purposes, will concern specifically the Lebesgue integral.

Riemann–Stieltjes Integral

Regarding the first generalization, let us observe that in the approximating sums $\underline{S}_n$ in the Riemann integral given by (7.7), we can write the terms $g_{n,k}^{(-)}(x_{n,k} - x_{n,k-1})$ as $g_{n,k}^{(-)}l([x_{n,k-1}, x_{n,k}])$, where $l(\cdot)$ stands for the length of the set (in this case, of a single interval). This is similar to the sums (7.8) approximating the Lebesgue integral.

Now, instead of the length being the difference between the coordinates of endpoints, one can take a “length-like” function, defined as the differences between values of some function, F , of coordinates. Thus, the terms of the approximating sums are now $g_{n,k}^{(-)}[F(x_{n,k}) - F(x_{n,k-1})]$. Naturally, function F has to satisfy some conditions if such an extension is to lead to a meaningful concept of an integral.

Without striving for a general definition (which can be found in most books on advanced calculus), we will simply consider the case where F is a cdf, meaning a nondecreasing function, continuous on the right, satisfying the conditions $\lim_{x \rightarrow -\infty} F(x) = 0$ and $\lim_{x \rightarrow \infty} F(x) = 1$.

The common limit (if it exists) of the two sequences

$$\underline{S}_n = \sum g_{n,k}^{(-)}[F(x_{n,k}) - F(x_{n,k-1})]$$

and

$$\bar{S}_n = \sum g_{n,k}^{(+)}[F(x_{n,k}) - F(x_{n,k-1})]$$

will be denoted $\int_a^b g(x) dF(x)$, and called the Riemann–Stieltjes (R–S) integral of function g with respect to function F . Again, we omit the details of an extension of the concept to the improper integral

$$\int_{-\infty}^{+\infty} g(x) dF(x).$$

■ EXAMPLE 7.13

Let us consider a special case where F_X is a cdf of a discrete random variable with the set of possible values $x_1, x_2, \dots$ and the corresponding probabilities $P\{X = x_i\} = p_i$. As we know from Chapter 5, the cdf of X is a step function that is constant between points x_i , and its steps equal p_i at points x_i .

The approximating (lower) sum $\underline{S}_n$ equals

$$\underline{S}_n = \sum g_{n,k}^{(-)}[F_X(x_{n,k}) - F_X(x_{n,k-1})].$$

Here the difference $F_X(x_{n,k}) - F_X(x_{n,k-1})$ of values of F_X at two consecutive points of the n th partition is zero if the interval $(x_{n,k-1}, x_{n,k}]$ does not contain any point x_i of the increase of F_X . When the partition becomes finer as $n \rightarrow \infty$, the only terms in $\underline{S}_n$ that remain will be those corresponding to intervals covering the points x_i , and the differences $F_X(x_{n,k}) - F_X(x_{n,k-1})$ for intervals covering point x_i will converge to p_i . Under some regularity assumptions (e.g., continuity of g), the values $g_{n,k}^{(-)}$ corresponding to nonzero terms will converge to corresponding values $g(x_i)$, and the limit will be

$$\int_{-\infty}^{+\infty} g(x) dF_X(x) = \sum_i g(x_i)p_i. \quad (7.9)$$

From formula (7.9) for the special case $g(x) = x$, we see that

$$\int_{-\infty}^{+\infty} x dF_X(x) = \sum x_i P\{X = x_i\},$$

provided that $\sum |x_i| P\{X = x_i\} < \infty$ (which turns out to be the condition for existence of the improper integral $\int_{-\infty}^{+\infty} x dF_X(x)$).

We have the following theorem:

Theorem 7.3.1 *If X is a discrete variable with cdf F_X and if $E(X)$ exists, then*

$$E(X) = \int_{-\infty}^{+\infty} x dF_X(x).$$

■ EXAMPLE 7.14

Consider the R–S integral in the case where the function F is a cdf of a continuous random variable so that $F'(x) = f(x)$, where f is the density of random variable X . In this case, the lower approximating sum can be written, using the mean value theorem, as

$$\underline{S}_n = \sum_k g_{n,k}^{(-)} [F(x_{n,k}) - F(x_{n,k-1})] = \sum_k g_{n,k}^{(-)} f(u_{n,k})(x_{n,k} - x_{n,k-1}),$$

where $u_{n,k}$ is a point between $x_{n,k-1}$ and $x_{n,k}$. Again, if g is continuous, the limiting value of $\underline{S}_n$ (and also $\overline{S}_n$) will be the integral of $g(x)f(x)$ between appropriate finite or infinite limits. Thus, in this case, the R–S integral becomes

$$\int g(x) dF(x) = \int g(x) f(x) dx.$$

Again, taking $g(x) = x$, we obtain

Theorem 7.3.2 *If X is a continuous random variable with cdf F_X , and if $E(X)$ exists, then*

$$E(X) = \int_{-\infty}^{+\infty} x dF_X(x).$$

We therefore have a single expression for an expected value of a random variable, which reduces to formulas (7.2) and (7.4) in the case of discrete and continuous random variables.

Lebesgue–Stieltjes Integral

The Lebesgue–Stieltjes integral is defined in very much the same way as the Lebesgue integral. If g is bounded (e.g., on interval $[a, b]$) so that $A \leq g(x) \leq B$ for some A and B , we use the sequence of partitions

$$A = y_{n,0} < y_{n,1} < \cdots < y_{n,2^n} = B$$

and take the (lower) approximating sum as

$$\underline{S}_n = \sum_{k=0}^{2^n} y_{n,k} l_F(C_{n,k}),$$

where $C_{n,k} = \{x : y_{n,k} \leq g(x) < y_{n,k+1}\}$, $k = 0, 1, \dots, 2^n - 1$, $C_{n,2^n} = \{x : g(x) = B\}$, and l_F is the generalized length of a set. This length is induced by the cdf F , in the sense explained above (see the construction of the Riemann–Stieltjes integral). The upper sum $\bar{S}_n$ is defined similarly, and the common limit (if it exists) of these two sequences of sums is the Lebesgue–Stieltjes integral

$$\int_a^b g(x) dF(x).$$

Again, we omit the details of the extension to integrals over an infinite interval.

As before, if the function g is R–S integrable with respect to F , then it is also L–S integrable, and the integrals are equal. If F corresponds to a discrete distribution with masses p_i at points x_i , then

$$\int_{-\infty}^{+\infty} g(x) dF(x) = \sum_i g(x_i) p_i,$$

provided that $\sum |g(x_i)| p_i < \infty$. If F is a cdf of continuous distribution with density f , then

$$\int_{-\infty}^{+\infty} g(x) dF(x) = \int_{-\infty}^{+\infty} g(x) f(x) dx,$$

provided that $\int_{-\infty}^{+\infty} |g(x)| f(x) dx < \infty$.

Lebesgue Integral: General Case

We now outline the second direction of generalization, applicable only to the case of the Lebesgue (or Lebesgue–Stieltjes) integral.

This generalization, by far the most important and profound extension of the concept of the Lebesgue integral, is based on the observation that approximating sums (7.8) make sense also for functions g that are defined on sets other than the real line. In fact, we can consider real-valued functions g defined on an arbitrary set (in particular, the sample space S). The sets $C_{n,k}$ are then subsets of sample space (hence events), and we let probability P play the role of length l .

In a more familiar notation, let X be a random variable defined on sample space S . Assume first that $X \geq 0$. Similarly to (7.8), we define the approximating sums as

$$S_n = \sum_{k=0}^{n2^n-1} \frac{k}{2^n} P \left\{ s : \frac{k}{2^n} \leq X(s) < \frac{k+1}{2^n} \right\}. \quad (7.10)$$

Observe that as n increases, the partitions become finer and also their range increases. Observe also that according to the comment made in the footnote on the opening page of Chapter 5, the probabilities in the sum (7.10) are well defined (i.e., arguments of probabilities are events, if X is a random variable).

We will omit the details of construction, which can be found in any advanced text on probability. Roughly, two properties are shown:

1. For every random variable $X \geq 0$, the sums S_n converge to a finite or infinite limit.
2. This limit exists and is the same if instead of $y_{n,k} = k/2^n$, we take any other sequence of partitions, provided that these partitions become finer and eventually cover the whole set $[0, \infty)$.

Consequently, we define the integral as the limit of sums (7.10):

$$\int_S X dP = \lim S_n.$$

The extension to arbitrary random variables (not necessarily nonnegative) now consists in defining

$$X^+ = \max(X, 0) \quad \text{and} \quad X^- = \max(-X, 0)$$

so that $X = X^+ - X^-$ is represented as a difference of two nonnegative functions. One then puts

$$\int_S X \, dP = \int_S X^+ \, dP - \int_S X^- \, dP,$$

provided that at least one of the integrals on the right-hand side is finite (so that only indeterminate case $\infty - \infty$ is ruled out).

We now have, more formally, the following definition:

Definition 7.3.1 The *expectation* $E(X)$ of a random variable X is defined as the Lebesgue integral of X :

$$E(X) = \int_S X \, dP,$$

provided $E|X| = \int_S |X| \, dP < \infty$. If $\int_S X^+ \, dP = \infty$ while we have $\int_S X^- \, dP < \infty$, we define $E(X) = \infty$ (and similarly for $E(X) = -\infty$). If $\int_S X^+ \, dP = \int_S X^- \, dP = \infty$, we say that expectation of X does not exist. $\square$

The expectation of a function of two (or more than two) random variables is defined in a similar way. To sketch the construction, consider a nonnegative function $g(X, Y)$ of a pair of random variables (X, Y) . The sums approximating the integral $E[g(X, Y)]$ are of the following form, similar to (7.10),

$$S_n = \sum_{k=0}^{n2^n-1} \frac{k}{2^n} P \left\{ s : \frac{k}{2^n} \leq g(X(s), Y(s)) < \frac{k+1}{2^n} \right\}.$$

The rest of the construction is analogous to the case of a single random variable, leading to the definition of the expectation $E[g(X, Y)]$.

7.4 PROPERTIES OF EXPECTATION

The identification of the expected value of a random variable X with Lebesgue integral of X allows us to formulate the properties of the latter as the properties of the expectation. Again, for the proofs, we refer the reader to any advanced textbook on probability.

Theorem 7.4.1 *The expectation of random variables has the following properties:*

(a) *Linearity.* If $E(X)$ exists, then for all α, β ,

$$E(\alpha X + \beta) = \alpha E(X) + \beta. \quad (7.11)$$

(b) *Nonnegativity.* If $X \geq 0$, then $E(X) \geq 0$.

(c) *Modulus inequality.* For any random variable X ,

$$|E(X)| \leq E(|X|).$$

We list here two of the most important consequences of this theorem. By putting $\beta = 0$ in (7.11), we have $E(\alpha X) = \alpha E(X)$. This means, in particular, that if we change the units of

measurement of X , then the expectation of X changes in the same way as X . On the other hand, by putting $\alpha = 0$, we obtain the property $E(\beta) = \beta$. So the expectation of a random variable equal to a constant β is equal to the same constant.

In the future, we will often use the following theorem, which gives a sufficient condition for the existence of an expected value.

Theorem 7.4.2 *If $|X| \leq Y$, where $E(Y) < \infty$, then $E(X)$ exists and is finite.*

Thus, to prove that the expectation of a random variable exists, it suffices to find another random variable Y that dominates it, and whose expectation exists. The domination $|X| \leq Y$ means that $|X(s)| \leq Y(s)$ for every point s in sample space $\mathcal{S}$. Similarly, the symbol $\lim_{n \rightarrow \infty} X_n = X$ is understood as a pointwise limit of random variables as functions on $\mathcal{S}$, that is, $\lim_{n \rightarrow \infty} X_n(s) = X(s)$ for every $s \in \mathcal{S}$.

Next, we give two principal theorems that connect expectation with convergence, allowing passing to the limit under the integral (expectation) sign.

Theorem 7.4.3 (Monotone Convergence Theorem) *If $X_1, X_2, \dots$ is a sequence of random variables such that $0 \leq X_1 \leq X_2 \leq \dots$ and*

$$\lim_{n \rightarrow \infty} X_n = X,$$

then

$$\lim_{n \rightarrow \infty} E(X_n) = E(X).$$

Theorem 7.4.4 (Dominated Convergence Theorem) *Let $X_1, X_2, \dots$ be a sequence of random variables satisfying the condition*

$$\lim_{n \rightarrow \infty} X_n = X.$$

Moreover, assume that there exists a random variable Y with $E(Y) < \infty$ such that $|X| \leq Y$ and $|X_n| \leq Y$ for $n = 1, 2, \dots$. Then

$$\lim_{n \rightarrow \infty} E(X_n) = E(X).$$

In both theorems, the assertion is that $\lim E(X_n) = E(\lim X_n)$, which means that we may interchange the order of integration and passage to the limit.

Instead of presenting formal proofs, we will show the necessity of the assumptions with examples.

■ EXAMPLE 7.15

We will now show by an example that the monotonicity alone is not enough for the assertion in the monotone convergence theorem; nonnegativity is essential. We let $\mathcal{S} = (0, 1)$ and let s be chosen according to the uniform distribution. We let X_n be defined as

$$X_n(s) = \begin{cases} -1/s & \text{if } 0 < s < 1/n \\ 0 & \text{if } 1/n \leq s < 1. \end{cases}$$

For each fixed s , the sequence $\{X_n(s)\}$ is of the form

$$-1/s, -1/s, \dots, -1/s, 0, 0, \dots,$$

hence is monotone, and $\lim X_n \equiv 0$. However, $EX_n = \int_0^{1/n} (-1/s) ds = -\infty$, while the integral of the limit random variable $X \equiv 0$ is $EX = 0$.

■ EXAMPLE 7.16

We will show that the existence of the bound Y with finite expectation is essential in the dominated convergence theorem. Again, let $S = (0, 1)$, with P being uniform on S . Now

$$X_n(s) = \begin{cases} 4n^2 s & \text{for } 0 < s < 1/(2n) \\ 4n(1 - ns) & \text{for } 1/(2n) \leq s < 1/n \\ 0 & \text{for } 1/n \leq s < 1. \end{cases}$$

For every fixed s , the sequence of numbers $\{X_n(s)\}$ converges to 0. Again, the limiting random variable is $X \equiv 0$, and $E(X) = 0$. But it is easy to check that $E(X_n) = 1$ for every n so that the assertion of the theorem does not hold. This time each random variable X_n is bounded, but there is no integrable common bound for all of them.

It is now necessary to connect two definitions of expectation of a random variable X : as a Lebesgue integral of X , and as a Riemann–Stieltjes integral of function $g(x) \equiv x$ with respect to cdf of X .

One may raise a doubt: Starting with Chapter 1, we stress the fact that for the same phenomenon (for the same random variable), the sample space S can be chosen in a number of different ways. But if we choose different sample spaces for describing the same random variable X , how can we guarantee that the integral of X is the same, regardless of the choice of S ?

The answer lies in the next theorem, which provides methods of computing Lebesgue integrals by reducing them to Lebesgue–Stieltjes and Riemann–Stieltjes integrals.

Theorem 7.4.5 *Let X be a random variable defined on probability space $\mathcal{S}$, and let $F(x)$ be its cdf. If $\int_{\mathcal{S}} |X| dP < \infty$, then*

$$\int_{\mathcal{S}} X dP = \int_{-\infty}^{+\infty} x dF(x). \quad (7.12)$$

More generally, if g is a real function such that $\int_{\mathcal{S}} |g(X)| dP < \infty$, then

$$E[g(X)] = \int_{\mathcal{S}} g(X) dP = \int_{-\infty}^{+\infty} g(x) dF(x). \quad (7.13)$$

The right-hand sides of formulas (7.12) and (7.13) provide means of computing the expectations. In the case of discrete and continuous random variables, formula (7.12) reduces to (7.2) or (7.4). Similarly, formula (7.13) reduces to

$$Eg(X) = \sum_i g(x_i) P\{X = x_i\} \quad (7.14)$$

and

$$Eg(X) = \int_{-\infty}^{+\infty} g(x) f(x) dx. \quad (7.15)$$

■ EXAMPLE 7.17

Formulas (7.12) and (7.13) are sometimes referred to as “theorems of the unconscious statistician.” The reason for the name is that in calculating expectations of some random variables, a statistician chooses either the left or right side of these formulas, often without being aware that using the other side may occasionally be simpler. We illustrate the situation by carrying out the calculations for both sides of (7.15).

Imagine that we have a stick of length 1, and we break it at a random point (i.e., the breaking point is chosen accordingly to the uniform distribution). What is the expected length of the longer of the two parts?

If we take the interval $[0, 1]$ as the sample space, with measure P being uniform (i.e., probabilities are proportional to lengths), then the length of the longer part of the stick, if the break occurs at s , is

$$X = \max(s, 1 - s).$$

The graph of X is presented at Figure 7.4(a). It is clear that $E(X) = \int_{\mathcal{S}} X dP = \int_0^1 \max(s, 1 - s) ds = 3/4$. This can be obtained by actually computing the integral or by observing that the area under the curve in Figure 7.4(a) is three quarters of the square of side 1.

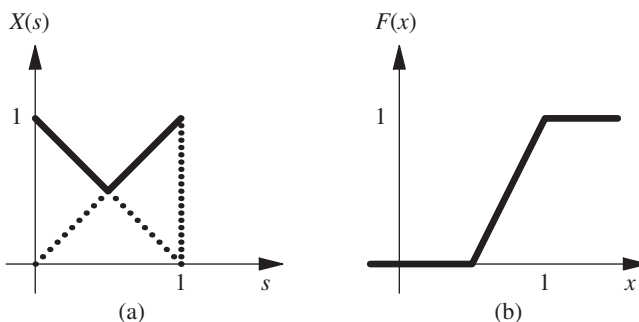


Figure 7.4 Graph of $X = \max(s, 1 - s)$ and its cdf.

We can also find the cdf of X and, realizing that X is a continuous random variable, use the right-hand side of (7.15). Since $1/2 \leq X \leq 1$, we can write

$$\begin{aligned} P\{X \leq x\} &= P\{\max(s, 1 - s) \leq x\} = P\{s \leq x, 1 - s \leq x\} \\ &= P\{1 - x \leq s \leq x\} = x - (1 - x) = 2x - 1. \end{aligned}$$

Consequently,

$$F(x) = \begin{cases} 0 & x \leq \frac{1}{2} \\ 2x - 1 & \frac{1}{2} \leq x \leq 1 \\ 1 & x \geq 1 \end{cases}$$

[see Figure 7.4(b)], and

$$E(X) = \int_{-\infty}^{+\infty} x dF(x) = \int_{1/2}^1 x dF(x) = \int_{1/2}^1 x 2 dx = \frac{3}{4}.$$

Let us now consider functions of two random variables—we will only discuss discrete and continuous bivariate distributions as the two most important cases. Let $g(x, y)$ be a real function such that $E|g(X, Y)| < \infty$. Suppose first that the distribution of (X, Y) is concentrated on $A \times B$, where $A = \{x_1, x_2, \dots\}$ and $B = \{y_1, y_2, \dots\}$. Let $p_{ij} = P\{X = x_i, Y = y_j\}$. Then

$$E[g(X, Y)] = \sum_{ij} g(x_i, y_j) p_{ij}. \quad (7.16)$$

Since $E|g(X, Y)| = \sum |g(x_i, y_j)|p_{ij} < \infty$, the sum (7.16) does not depend on the order of summation. So one can choose the order that leads to simpler calculations:

$$\begin{aligned} E[g(X, Y)] &= \sum_{i=1}^{\infty} \sum_{j=1}^{\infty} g(x_i, y_j) p_{ij} = \sum_{j=1}^{\infty} \sum_{i=1}^{\infty} g(x_i, y_j) p_{ij} \\ &= \sum_{k=1}^{\infty} \sum_{r=1}^k g(x_r, y_{k-r}) p_{r, k-r}, \end{aligned}$$

and so on. The choice depends on the function g and probabilities p_{ij} .

Let (X, Y) have the joint density $f(x, y)$. We then have

Theorem 7.4.6 *If $E|g(X, Y)| < \infty$, then*

$$\begin{aligned} E[g(X, Y)] &= \int_{R^2} \int g(x, y) f(x, y) dx dy \\ &= \int_{-\infty}^{+\infty} \left[\int_{-\infty}^{+\infty} g(x, y) f(x, y) dx \right] dy \\ &= \int_{-\infty}^{+\infty} \left[\int_{-\infty}^{+\infty} g(x, y) f(x, y) dy \right] dx. \end{aligned} \tag{7.17}$$

The last two expressions provide the computational formulas that can be used in practice, since they reduce the computation of a double (two-dimensional) integral to an iteration of two single (one-dimensional) integrals.

As an illustration of applicability of Theorem 7.4.6, we give the proof (in the case of continuous random variables) of the following theorem:

Theorem 7.4.7 *Assume that $E|X| < \infty$ and $E|Y| < \infty$. Then*

$$E(X + Y) = E(X) + E(Y).$$

Proof: Let $f(x, y)$ be the density of (X, Y) , and let f_X and f_Y be the marginal densities of X and Y , respectively. Letting $g(x, y) = x + y$ in (7.17), we have

$$\begin{aligned} E(X + Y) &= \int_{R^2} \int (x + y) f(x, y) dx dy \\ &= \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} x f(x, y) dx dy + \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} y f(x, y) dx dy \\ &= \int_{-\infty}^{+\infty} x \left[\int_{-\infty}^{+\infty} f(x, y) dy \right] dx + \int_{-\infty}^{+\infty} y \left[\int_{-\infty}^{+\infty} f(x, y) dx \right] dy \\ &= \int_{-\infty}^{+\infty} x f_X(x) dx + \int_{-\infty}^{+\infty} y f_Y(y) dy = E(X) + E(Y). \end{aligned}$$

The order of integration can be changed since we assumed that $\int_{-\infty}^{+\infty} |x| f_X(x) dx < \infty$, $\int_{-\infty}^{+\infty} |y| f_Y(y) dy < \infty$. □

The property of additivity extends immediately to any finite number of random variables with finite expectations:

$$E(X_1 + \cdots + X_n) = E(X_1) + \cdots + E(X_n),$$

or, combining it with Theorem 7.4.1(a), we have

$$E(\beta_1 X_1 + \cdots + \beta_n X_n + \alpha) = \sum_{i=1}^n \beta_i E(X_i) + \alpha.$$

Finally, the nonnegativity property (b) in Theorem 7.4.1 implies that expectation is monotone: if X and Z have finite expectations, then $Y \leq Z$ implies $E(Y) \leq E(Z)$. To see this, we write $Z = Y + (Z - Y)$, where now $Z - Y \geq 0$. We have

$$E(Z) = E(Y) + E(Z - Y) \geq E(Y).$$

We have also the following important theorem:

Theorem 7.4.8 *If X and Y are independent random variables such that $E|XY| < \infty$, then*

$$E(XY) = E(X)E(Y).$$

Proof: We give the proof only for the case of continuous random variables.

$$\begin{aligned} E(XY) &= \int_{\mathbb{R}^2} \int xyf(x, y) dx dy = \int_{\mathbb{R}^2} \int xyf_X(x)f_Y(y) dx dy \\ &= \left(\int_{-\infty}^{+\infty} xf_X(x) dx \right) \times \left(\int_{-\infty}^{+\infty} yf_Y(y) dy \right) = E(X)E(Y). \end{aligned} \quad \square$$

Replacing the double integral by the iterated integrals, as well as the change of the order of integration (or summation), is very often taken for granted. Actually, the fact that iterated integrals are equal to one another is not a “law of nature,” as some are inclined to believe. It is a fact that is true under specific assumptions, namely under the existence of the double integral.

We will not provide here the precise statement of the relevant theorem (see the Fubini theorem or Lebesgue integration in any advanced textbook on probability). Instead, we will give a simple example that shows that the iterated integrals need not be equal.

■ EXAMPLE 7.18

Consider $f(x, y)$ described in Figure 7.5. Obviously, here $\int_0^\infty f(x, y) dx = 0$ for every y ,

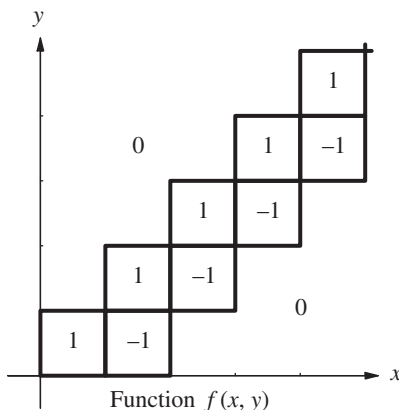


Figure 7.5 Nonintegrable function whose iterated integrals exist and are not equal.

so $\int_0^\infty \int_0^\infty f(x, y) dx dy = 0$. On the other hand, $\int_0^\infty f(x, y) dy$ is 1 for $0 \leq x \leq 1$ and 0 otherwise, so $\int_0^\infty \int_0^\infty f(x, y) dx dy = 1$.

It ought to be clear that the reason why there is a difference between the values of two iterated integrals is that the double integral (the sum of infinitely many “volumes” $+1$ and infinitely many “volumes” -1) does not exist.

■ EXAMPLE 7.19

Consider now a function $h(x, y)$ similar to that of Example 7.18 given in Figure 7.6.

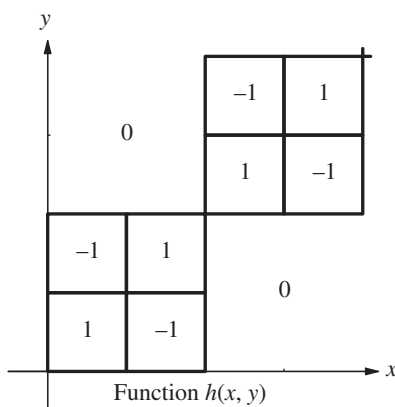


Figure 7.6 Nonintegrable function whose iterated integrals exist and are equal.

In this case, we have $\int_0^\infty h(x, y) dx = \int_0^\infty h(x, y) dy = 0$, so both iterated integrals exist and are equal zero. Still the double integral does not exist, for the same reason as in Example 7.18.

Examples 7.18 and 7.19 show that if the double integral does not exist, we cannot say anything about equality of iterated integrals.

PROBLEMS

7.4.1 Show that $E(X - E(X)) = 0$.

7.4.2 Assume that X has density $f(x) = ax + bx^3$ for $0 \leq x \leq 2$ and $f(x) = 0$ otherwise. Find a and b if $E(X^2) = 2.5$.

7.4.3 Find $E(X)$, $E(1/X)$, and $E(2^X)$ if $X = 1, 2, \dots, 8$ with equal probabilities $1/8$.

7.4.4 Let X be a random variable with density $f(x) = 1/2$ for $-1 \leq x \leq 1$ and $f(x) = 0$ otherwise. Find: **(i)** $E(X)$. **(ii)** $E(X^2)$. **(iii)** $E(2X - 3)^2$.

7.4.5 Let X have the density

$$f(x) = \begin{cases} cx & 0 \leq x \leq 1 \\ c(2 - x) & 1 \leq x \leq 2 \\ 0 & \text{otherwise.} \end{cases}$$

Find: **(i)** c . **(ii)** $E(X)$. **(iii)** $E(2 - X)^3$. **(iv)** $E[1/(2 - X)]$.

7.4.6 An urn contains w white and r red balls. We draw $n \leq r$ balls from the urn without replacement, and we let X be the number of red balls drawn. Find: $E(X)$, by defining indicator variables: **(i)** $X_1, \dots, X_n$ such that $X = X_1 + \dots + X_n$. **(ii)** $Y_1, \dots, Y_r$ such that $X = Y_1 + \dots + Y_r$.

7.4.7 A cereal company puts a plastic bear in each box of cereal. Every fifth bear is red. If you have three red bears, you get a free box of the cereal. If you decide to keep buying this cereal until you get one box free, how many boxes would you expect to buy before getting a free one? [Hint: Represent the answer as $E(X_1 + X_2 + X_3)$, where X_i is the number of boxes you buy after getting the $(i-1)$ st red bear and until getting i th red bear.]

7.4.8 Show that if X is such that $P(a \leq X \leq b) = 1$, then $E(X)$ exists and $a \leq E(X) \leq b$.

7.4.9 We say that X is *stochastically smaller* than Y ($X \leq_{st} Y$) if $P\{X \leq t\} \geq P\{Y \leq t\}$ for all t . Show that if X and Y have finite expectations and $X \leq_{st} Y$, then $E(X) \leq E(Y)$. (Hint: Start with nonnegative X and Y and use Theorem 7.2.2. Then use the decomposition into a positive and negative part.) Show also that the converse assertion is false: there exist random variables X and Y such that $E(X) < E(Y)$ and X is not stochastically smaller than Y .

7.5 MOMENTS

We begin with the following definition:

Definition 7.5.1 For any random variable X , the expectation of X^n , if it exists, will be called the n th *ordinary moment* (or *the moment of order n*) of X and denoted

$$m_n = E(X^n). \quad \square$$

The n th moment, m_n , exists if $E|X|^n < \infty$. The moment of the order 0 always exists and equals 1, while m_1 is simply the expectation of X . If the n th moment exists, it may be computed from the formula

$$m_n = E(X^n) = \int_{-\infty}^{+\infty} x^n dF(x),$$

where F is the cdf of X ; this formula is obtained by substituting $g(x) = x^n$ in (7.13). Observe that any bounded random variable X , that is, a random variable such that $P(|X| \leq M) = 1$ for some $M < \infty$, has finite moments of any order. This follows at once from Theorem 7.4.2, and in this case, we have $m_n \leq M^n$.

■ EXAMPLE 7.20

Let X have $\text{POI}(\lambda)$ distribution. Then

$$m_1 = E(X) = \sum_{k=0}^{\infty} k \frac{\lambda^k}{k!} e^{-\lambda} = \lambda e^{-\lambda} \sum_{k=1}^{\infty} \frac{\lambda^{k-1}}{(k-1)!} = \lambda.$$

On the other hand,

$$\begin{aligned} m_2 = E(X^2) &= \sum_{k=0}^{\infty} k^2 \frac{\lambda^k}{k!} e^{-\lambda} = \sum_{k=1}^{\infty} [k(k-1) + k] \frac{\lambda^k}{k!} e^{-\lambda} \\ &= \lambda^2 \sum_{k=2}^{\infty} e^{-\lambda} \frac{\lambda^{k-2}}{(k-2)!} + \lambda \sum_{k=1}^{\infty} e^{-\lambda} \frac{\lambda^{k-1}}{(k-1)!} = \lambda^2 + \lambda, \end{aligned}$$

since the second sum equals $m_1 = \lambda$.

■ **EXAMPLE 7.21**

Let X have the $U[a, b]$ distribution. Then its density is $f(x) = 1/(b - a)$ for $a \leq x \leq b$ and $f(x) = 0$ otherwise so that

$$m_n = \int_a^b x^n \times \frac{dx}{b-a} = \frac{1}{n+1} \times \frac{b^{n+1} - a^{n+1}}{b-a}.$$

When the distribution is symmetric around 0, then $a = -b$, and we have $m_{2k} = b^{2k}/(2k+1)$ while $m_{2k+1} = 0$.

Before introducing the concepts which will illustrate the usefulness of the notion of moments, let us introduce some definitions pertaining to a variety of types of moments.

Definition 7.5.2 An *absolute moment* of order n is defined as

$$\beta_n = E(|X|^n). \quad \square$$

The existence of an absolute moment of a given order implies the existence of an “ordinary” moment m_n of the same order. Note that the order of absolute moments need not be an integer. The same applies to ordinary moments of positive random variables.

Definition 7.5.3 Ordinary moments of the random variable $Y = X - E(X) = X - m_1$, are called *central moments* of X so that

$$\gamma_n = E[(X - m_1)^n]. \quad \square$$

Clearly, the first central moment γ_1 is always equal to 0.

Definition 7.5.4 A *factorial moment* of X of order n is defined as

$$\pi_n = E[X(X-1) \cdots (X-n+1)]. \quad \square$$

We will now prove the following:

Theorem 7.5.1 If an absolute moment of order $\alpha > 0$ exists, then all moments (ordinary, absolute, central, and factorial) of orders $r \leq \alpha$ exist.

Proof: We will prove first that if $E(|X|^\alpha) < \infty$, then $E(|X|^r) < \infty$ for all $r < \alpha$. Clearly, if $|X| \geq 1$, then $|X|^r \leq |X|^\alpha$, while if $|X| < 1$, then $|X|^r \leq 1$. Consequently,

$$|X|^r \leq \max(1, |X|^\alpha), \quad (7.18)$$

and (see Theorem 7.4.2) it remains to show that the right-hand side of (7.18) has finite expectation.

Now, if $E|X|^\alpha < \infty$, then

$$E(|X|^\alpha) = \int_{-1}^1 |x|^\alpha dF(x) + \int_{|x| \geq 1} |x|^\alpha dF(x),$$

and it follows that $\int_{|x| \geq 1} |x|^\alpha dF(x) < \infty$ as a difference of two finite quantities. Thus, we can write

$$E(|X|^r) \leq E\{\max(1, |X|^\alpha)\} = \int_{-1}^1 dF(x) + \int_{|x| \geq 1} |x|^\alpha dF(x) < \infty.$$

It remains to prove that if $E|X|^\alpha < \infty$, then all other types of moments of order α exist. This is true for ordinary moments (by definition). Regarding central and factorial moments of order n , they are linear combinations of ordinary moments of order $k \leq n$, which proves the theorem. $\square$

The following theorem will be given without proof:

Theorem 7.5.2 (Liapunov Inequality) *If $0 < \alpha < \beta < \infty$, then*

$$\{E(|X|^\alpha)\}^{1/\alpha} \leq \{E(|X|^\beta)\}^{1/\beta}.$$

Note that the first part of the proof of Theorem 7.5.1 is an immediate consequence of the Liapunov inequality.

We will now introduce a function that will be a valuable tool in analyzing random variables and limiting behavior of their sums.

Definition 7.5.5 The function of real variable t defined as

$$m_X(t) = E(e^{tX})$$

is called the moment generating function (mgf) of a random variable X . □

For any random variable, its mgf exists for $t = 0$. If X is a positive random variable, and the mgf of X exists for some t_0 , then $m_X(t)$ exists for all $t \leq t_0$. This fact follows from Theorem 7.4.2.

■ EXAMPLE 7.22

If X has the $\text{BIN}(n, p)$ distribution, then

$$m_X(t) = \sum_{k=0}^n e^{tk} \binom{n}{k} p^k q^{n-k} = (pe^t + q)^n$$

so that $m_X(t)$ exists for all t .

■ EXAMPLE 7.23

If X has $\text{EXP}(\lambda)$ distribution, then we can write

$$m_X(t) = \int_0^\infty e^{tx} \times \lambda e^{-\lambda x} dx = \lambda \int_0^\infty e^{-(\lambda-t)x} dx = \frac{\lambda}{\lambda-t}, \quad (7.19)$$

provided that $t < \lambda$. For $t \geq \lambda$, the integral in (7.19) diverges.

Let us explore some of the properties of moment generating functions. First, observe that the concept of the mgf is connected with a distribution rather than with a random variable: two different random variables with the same distribution will have the same mgf.

The name “moment generating function” is related to the following theorem:

Theorem 7.5.3 *Let X be a random variable with the mgf $m_X(t)$, assumed to exist in some neighborhood of $t = 0$. If $E(|X|^n) < \infty$, then for $k = 1, \dots, n$, the k th moment of X is given by the formula*

$$m_k = \left. \frac{d^k}{dt^k} m_X(t) \right|_{t=0}.$$

Proof: Differentiating formally the expression for the mgf under the sign of expectation (i.e., under the integral or summation sign), we obtain

$$m'_X(t) = E(Xe^{tX}), \quad m''_X(t) = E(X^2e^{tX}), \quad \dots, \quad \frac{d^k}{dt^k}m_X(t) = E(X^k e^{tX}),$$

for $k = 1, 2, \dots, n$. Substitution of $t = 0$ gives the required result.

The validity of this argument depends crucially on whether or not formal differentiation under the integral sign is allowed. The answer is positive if the corresponding derivatives are absolutely integrable for $t = 0$, which is ensured by the existence of the n th absolute moment. $\square$

■ EXAMPLE 7.24

Since the mgf of random variable with $\text{BIN}(n, p)$ distribution is $m_X(t) = (pe^t + q)^n$, we have $m'_X(t) = n(pe^t + q)^{n-1}pe^t$ while $m''_X(t) = n(n-1)(pe^t + q)^{n-2}(pe^t)^2 + n(pe^t + q)^{n-1}pe^t$. Thus $E(X) = m'_X(0) = np$ and $E(X^2) = m''_X(0) = n(n-1)p^2 + np$.

■ EXAMPLE 7.25

The mgf of the $U[a, b]$ distribution is

$$m_X(t) = \int_a^b e^{tx} \times \frac{1}{b-a} dx = \frac{e^{bt} - e^{at}}{t(b-a)}. \quad (7.20)$$

Determination of the values of derivatives of $m_X(t)$ at $t = 0$ requires repeated usage of the de L'Hospital rule. We can, however, expand exponentials into power series, and after some algebra, we obtain

$$m_X(t) = 1 + \frac{1}{2!} \frac{b^2 - a^2}{b-a} t + \frac{1}{3!} \frac{b^3 - a^3}{b-a} t^2 + \dots$$

This is a Taylor expansion of $m_X(t)$ about $t = 0$. So we must have

$$\begin{aligned} m_n &= m_X^{(n)}(0) = \frac{n!}{(n+1)!} \frac{b^{n+1} - a^{n+1}}{b-a} \\ &= \frac{1}{n+1} [b^n + b^{n-1}a + b^{n-2}a^2 + \dots + a^n]. \end{aligned}$$

Next, we will prove

Theorem 7.5.4 *If X is a random variable with mgf $m_X(t)$, then the random variable $Y = \alpha X + \beta$ has the mgf*

$$m_Y(t) = e^{\beta t} m_X(\alpha t). \quad (7.21)$$

Proof: From the properties of the expectation,

$$m_Y(t) = E(e^{tY}) = E(e^{t(\alpha X + \beta)}) = E(e^{(\alpha t)X} \times e^{\beta t}) = e^{\beta t} m_X(\alpha t). \quad \square$$

■ **EXAMPLE 7.26**

Let us find the mgf's of random variables U and V , with distributions $U[0, 1]$ and $U[-1, 1]$, respectively.

If variable X has a $U[a, b]$ distribution, and hence has the mgf given by (7.20), then $U = (X - a)/(b - a)$ has a distribution uniform on $[0, 1]$. Using formulas (7.20) and (7.21) for $\alpha = 1/(b - a)$ and $\beta = -a/(b - a)$, we obtain

$$m_U(t) = e^{-at/(b-a)} m_X\left(\frac{t}{b-a}\right) = \frac{e^t - 1}{t}. \quad (7.22)$$

Next, if V has a $U[-1, 1]$ distribution, then $V = 2U - 1$. Taking $\alpha = 2$ and $\beta = -1$ and using (7.22), we get

$$m_V(t) = e^{-t} \times \frac{e^{2t} - 1}{2t} = \frac{e^t - e^{-t}}{2t} = \frac{\sinh t}{t}.$$

Obviously, the expressions for these mgf's can be obtained directly from the definition.

■ **EXAMPLE 7.27**

The mgf of a $N(\mu, \sigma^2)$ distribution can be found by first finding the mgf of a standard normal random variable $Z \sim N(0, 1)$ as

$$m_Z(t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{tz} e^{-\frac{z^2}{2}} dz = \frac{e^{t^2/2}}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-(1/2)(z-t)^2} dz = e^{t^2/2}.$$

Then, for any $X \sim N(\mu, \sigma^2)$, $X = \mu + \sigma Z$, and based on Theorem 7.5.4 we obtain

$$m_X(t) = e^{\mu t + \sigma^2 t^2/2}. \quad (7.23)$$

We now prove

Theorem 7.5.5 *If X and Y are independent random variables with mgf's $m_X(t)$ and $m_Y(t)$, respectively, then the mgf of $X + Y$ is*

$$m_{X+Y}(t) = m_X(t)m_Y(t).$$

Proof: Observe that random variables e^{tX} and e^{tY} are independent for each t , so by Theorem 7.4.8, we have

$$m_{X+Y}(t) = E(e^{t(X+Y)}) = E(e^{tX})E(e^{tY}) = m_X(t)m_Y(t). \quad \square$$

Before proceeding with examples, we state one more important theorem. Its proof is beyond the scope of this book.

Theorem 7.5.6 *If X and Y are two random variables such that their mgf's $m_X(t)$ and $m_Y(t)$ coincide in some neighborhood of the point $t = 0$, then X and Y have the same distribution.*

Theorem 7.5.6 asserts that if two mgf's agree in some neighborhood of $t = 0$, then they agree for all t (for which they are defined), and that an mgf determines uniquely the distribution. In other words, random variables with different distributions must have different mgf's.

■ **EXAMPLE 7.28**

Let X and Y be independent and have Poisson distributions with means λ_1 and λ_2 , respectively. Let us first determine the mgf's of X and of Y . We have

$$m_X(t) = \sum_{k=0}^{\infty} e^{tk} \frac{\lambda_1^k}{k!} e^{-\lambda_1} = e^{-\lambda_1} \sum_{k=0}^{\infty} \frac{(\lambda_1 e^t)^k}{k!} = e^{\lambda_1(e^t - 1)}.$$

Similarly, $m_Y(t) = e^{\lambda_2(e^t - 1)}$, and by Theorem 7.5.6,

$$m_{X+Y}(t) = m_X(t)m_Y(t) = e^{(\lambda_1 + \lambda_2)(e^t - 1)}. \quad (7.24)$$

We recognize (7.24) as the mgf of the Poisson distribution. In view of Theorem 7.5.6, $X + Y$ has a Poisson distribution with mean $\lambda_1 + \lambda_2$.

The main disadvantage of mgf's is that they may not exist for any $t \neq 0$. There exist random variables X such that the random variable e^{tX} has no expectation for any $t \neq 0$. This restricts the usefulness of mgf's as a tool (to be explored in Chapter 9) for obtaining limit theorems. For proofs using mgf's to be valid, they have to be confined to classes of random variables for which the mgf's exist in some neighborhood of $t = 0$. The corresponding theorems, however, are typically valid without this assumption, and the proofs usually require nothing more than replacing the mgf's with so-called *characteristic functions* (chf's). Although we will not use chf's in this book, it is worthwhile to provide their definition and simplest properties. The concepts below require a rudimentary knowledge of complex numbers.

Definition 7.5.6 Let X be a random variable with the cdf F . The function φ_X of real argument t ($-\infty < t < +\infty$) defined as

$$\varphi_X(t) = E(e^{itX}) = \int e^{itx} dF(x) \quad (7.25)$$

is called the chf of X (or of the cdf F). □

In the definition above, i is the imaginary unit (i.e., $i^2 = -1$). From the formula $e^{i\xi} = \cos \xi + i \sin \xi$, we obtain

$$\varphi_X(t) = E\{\cos(tX)\} + iE\{\sin(tX)\}, \quad (7.26)$$

so $\varphi_X(t)$ is a complex-valued function of a real argument. Since $|e^{itX}| = \cos^2(tX) + \sin^2(tX) = 1$, the expectation in (7.25) exists for all t , so the chf's always exist.

Below we list some basic properties of chf's; the proofs are left as exercises. In particular, Theorems 7.5.3, 7.5.4, and 7.5.5 carry over to the case of chf's almost without any change. In the following theorems, $\varphi_X, \varphi_Y, \dots$ are chf's of random variables $X, Y, \dots$:

Theorem 7.5.7 For every random variable X , its chf $\varphi_X(t)$ is uniformly continuous and satisfies the conditions

$$\varphi_X(0) = 1, \quad |\varphi_X(t)| \leq 1, \quad (7.27)$$

for every real t . Moreover, for any real a, b ,

$$\varphi_{aX+b}(t) = \varphi_X(at)e^{ibt}. \quad (7.28)$$

In particular, for $a = -1, b = 0$, using (7.26,) we obtain

$$\varphi_{-X}(t) = \varphi(-t) = \overline{\varphi(t)}, \quad (7.29)$$

where $\bar{z}$ stands for the complex conjugate of z .

■ **EXAMPLE 7.29**

If $X \sim \text{BIN}(n, p)$, then

$$\varphi_X(t) = E(e^{itX}) = \sum_{k=0}^n e^{itk} \binom{n}{k} p^k q^n = (pe^{it} + q)^n = [(p \cos t + q) + ip \sin t]^n.$$

In this case $\varphi_X(t)$ is a periodic function.

■ **EXAMPLE 7.30**

If X is uniform on $[0, 1]$, then

$$\varphi_X(t) = \int_0^1 e^{itx} dx = \frac{1}{it}(e^{it} - 1).$$

If $Y = 2X - 1$, then Y has uniform distribution on $[-1, 1]$ and (7.28) gives

$$\varphi_Y(t) = \varphi_{2X-1}(t) = \varphi_X(2t)e^{-it} = \frac{1}{2it}(e^{2it} - 1)e^{-it} = \frac{\sin t}{t}.$$

■ **EXAMPLE 7.31**

If $X \sim N(0, 1)$, then

$$\varphi_X(t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} e^{itx} e^{-x^2/2} dx = \frac{e^{-t^2/2}}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} e^{-(1/2)(x-it)^2} dx = e^{-t^2/2}.$$

The addition of independent random variables corresponds to multiplications of their chf's.

Theorem 7.5.8 *If random variables X and Y are independent, then*

$$\varphi_{X+Y}(t) = \varphi_X(t)\varphi_Y(t).$$

As in the case of mgf's, chf's uniquely determine the distributions:

Theorem 7.5.9 *If two cdf's F and G have the same chf, then $F \equiv G$.*

Finally, the relationship between moments of X and behavior of chf in the neighborhood of $t = 0$ is given by the following theorem:

Theorem 7.5.10 *If X is a random variable such that $E(|X|^k) < \infty$ for some integer $k \geq 1$, then*

$$\varphi_X(t) = \sum_{j=0}^k \frac{i^j}{j!} m_j t^j + o(|t|^k)$$

in some neighborhood of $t = 0$. Here $m_j = E(X^j)$, and $o(x)$ is such that $\lim_{x \downarrow 0} o(x)/x = 0$.

PROBLEMS

- 7.5.1** Find the mgf of a discrete random variable X with distribution $P\{X = k\} = 1/n, k = 0, 1, \dots, n-1$.
- 7.5.2** Let X be a nonnegative integer-valued random variable. The function $g_X(s) = Es^X$, defined for $|s| \leq 1$, is called a *probability generating function*, or simply a generating function, of X . Find $g_X(s)$ for random variables with: **(i)** Geometric distribution. **(ii)** Binomial distribution. **(iii)** Poisson distribution.
- 7.5.3** Find the fourth factorial moment of the random variable X with a $\text{POI}(\lambda)$ distribution.
- 7.5.4** Find the mgf for a random variable with a density: **(i)** $f(x) = xe^{-x}$ for $x > 0$. **(ii)** $f(x; \sigma^2) = \sqrt{2/\pi}e^{-x^2/2\sigma^2}$ for $x > 0$.
- 7.5.5** Let $X_1, \dots, X_n$ be independent random variables with the same distribution $N(\mu, \sigma^2)$. Find constants α_n and β_n such that $U = (X_1 + \dots + X_n - \alpha_n)/\beta_n$ and X_1 have the same distribution.
- 7.5.6** A continuous random variable X is called *symmetric about c* if its density f satisfies the condition $f(c-x) = f(c+x)$ for all x . Show that: **(i)** If X is symmetric about c and $E(X)$ exists, then $E(X) = c$. **(ii)** If X is symmetric about 0, then all moments of odd order (if they exist) are equal to 0.
- 7.5.7** Let X be a random variable with $E(X) = \mu$, $E(X - \mu)^2 = \sigma^2$ and such that the third central moment $\gamma_3 = E(X - \mu)^3$ exists. The ratio γ_3/σ^3 is called the *coefficient of skewness*. Find skewness of the following distributions: **(i)** $U[0, 1]$. **(ii)** $f(x) = \alpha x^{\alpha-1}$ for $0 \leq x \leq 1$ and $f(x) = 0$ otherwise ($\alpha > 1$). **(iii)** $\text{BIN}(1, p)$. **(iv)** $\text{POI}(\lambda)$.
- 7.5.8** Let X be a random variable with $E(X) = \mu$, $E(X - \mu)^2 = \sigma^2$ and such that $\gamma_4 = E(X - \mu)^4$ exists. Then γ_4/σ^4 is called the *coefficient of kurtosis*. Find kurtosis of the following distributions: **(i)** $N(0, 1)$. **(ii)** $N(\mu, \sigma^2)$. **(iii)** $\text{BIN}(1, p)$. **(iv)** $\text{POI}(\lambda)$.
- 7.5.9** Let random variable X have mgf $m_X(t) = 0.2 + 0.4e^{-t} + 0.1e^t + 0.3e^{2t}$. Use m_X to obtain $E[X(X-1)(X-2)]$ —the third factorial moment of X .
- 7.5.10** Find the chf of the following distributions: **(i)** $\text{POI}(\lambda)$. **(ii)** $\text{GEO}(p)$. **(iii)** $\text{EXP}(\lambda)$. **(iv)** $N(\mu, \sigma^2)$.
- 7.5.11** A family $\mathcal{G}$ of distributions is said to be *closed under convolution*, if whenever independent random variables X and Y have distributions in $\mathcal{G}$; the same is true for the random variable $X + Y$. Show closeness under convolution in families of: **(i)** Poisson distributions; **(ii)** Normal distributions.
- 7.5.12** Show that if $\varphi(t)$ is a chf, then $|\varphi(t)|^2$ is also a chf.
- 7.5.13** Show that the distribution of X is symmetric around 0 if and only if $\varphi_X(t)$ is real.

7.6 VARIANCE

We will now introduce a concept that plays an extremely important role in statistical analysis.

Definition 7.6.1 If $E(X^2) < \infty$, then the second central moment of X ,

$$\text{Var}(X) = E[(X - m_1)^2],$$

is called the *variance* of X . Its positive square root is called the *standard deviation* of X . $\square$

The other symbols used for the variance are $V(X)$, σ_X^2 , and σ^2 .

We begin by listing some basic properties of the variance and providing preliminary examples. First, observe that

$$\begin{aligned}\text{Var}(X) &= E(X^2 - 2m_1X + m_1^2) = E(X^2) - 2m_1E(X) + m_1^2 \\ &= E(X^2) - [E(X)]^2.\end{aligned}\tag{7.30}$$

Formula (7.30) gives an alternative way of calculating the variance. Note that variance is always nonnegative, as the expectation of a nonnegative random variable $(X - m_1)^2$. Consequently, $E(X^2) \geq [E(X)]^2$; that is, $E(X^2)^{1/2} \geq |E(X)|$, which is a special case of the Liapunov inequality given in Theorem 7.5.2.

Theorem 7.6.1 *If $\text{Var}(X)$ exists, then*

$$\text{Var}(aX + b) = a^2\text{Var}(X).\tag{7.31}$$

Proof: Using the fact that $E(aX + b) = aE(X) + b$, we write

$$\begin{aligned}\text{Var}(aX + b) &= E[(aX + b)^2] - [aE(X) + b]^2 \\ &= E[a^2X^2 + 2abX + b^2] - a^2[E(X)]^2 - 2abE(X) - b^2 \\ &= a^2[E(X^2) - (E(X))^2] = a^2\text{Var}(X).\end{aligned}\quad \square$$

■ EXAMPLE 7.32

Let X have the Poisson distribution with mean λ . Then its mgf is $m_X(t) = e^{\lambda(e^t - 1)}$ (see Example 7.28). After some algebra, we obtain

$$m_X''(t) = \lambda(\lambda e^t + 1)e^{\lambda(e^t - 1) + t}$$

so that $E(X^2) = m_X''(0) = \lambda^2 + \lambda$. Now, $\text{Var}(X) = (\lambda^2 + \lambda) - \lambda^2 = \lambda$. Thus, for the Poisson distribution, the mean and variance coincide.

■ EXAMPLE 7.33

From Example 7.24, we know that if X has $\text{BIN}(n, p)$ distribution, then $E(X) = np$, $E(X^2) = n(n-1)p^2 + np$. Consequently, using (7.30), we obtain

$$\text{Var}(X) = n(n-1)p^2 + np - (np)^2 = np(1-p) = npq.\tag{7.32}$$

■ EXAMPLE 7.34

Let us find the variance of random variable X with a $U[a, b]$ distribution. We have $f(x) = 1/(b-a)$ for $a \leq x \leq b$, $E(X) = (a+b)/2$, and

$$E(X^2) = \frac{1}{3} \times \frac{b^3 - a^3}{b-a} = \frac{1}{3}[b^2 + ab + a^2],$$

so $\text{Var}(X) = (b-a)^2/12$.

■ EXAMPLE 7.35

In Example 7.23, we found that the mgf of the $\text{EXP}(\lambda)$ distribution is $m(t) = \lambda(\lambda - t)^{-1}$. Consequently, $m'(t) = \lambda(\lambda - t)^{-2}$, $m''(t) = 2\lambda(\lambda - t)^{-3}$. Then $E(X) = 1/\lambda$, $E(X^2) = 2/\lambda^2$ and $\text{Var}(X) = 1/\lambda^2$, so in the exponential distribution the mean and standard deviation coincide.

To interpret the variance, let us now consider the case of a discrete random variable. The variance equals

$$\text{Var}(X) = \sum_i (x_i - m_1)^2 P(X = x_i). \quad (7.33)$$

For the variance to be small, all terms of the sum (7.33) must be small. Hence, the values x_i with the large difference $|x_i - m_1|$ must have a low probability. Qualitatively speaking, a small variance means that the values of X are concentrated closely to the mean of X , so large deviations are unlikely. The following theorem provides another interpretation of the variance:

Theorem 7.6.2 *The mean square deviation from ξ , $E(X - \xi)^2$, is minimized at $\xi = E(X)$, and the minimal value equals $\text{Var}(X)$.*

Proof: Since $f(\xi) = E(X^2) - 2\xi E(X) + \xi^2$ represents a parabola with branches directed upward, the minimum occurs at the point ξ^* at which $f'(\xi^*) = 0$, so $\xi^* = E(X)$. $\square$

The function $f(\xi)$ corresponds to the case where, qualitatively speaking, “small errors are almost negligible, large errors are very serious.” The function $g(\xi)$ in Example 7.36 treats the seriousness of an error as proportional to its size.

■ EXAMPLE 7.36

A natural question that arises from problem posed in Theorem 7.6.2 is to determine ξ that minimizes

$$g(\xi) = E|X - \xi|.$$

The minimization of $g(\xi)$ is not as simple as that of $f(\xi)$. We have to turn here to the following result:

Theorem 7.6.3 *The mean absolute deviation from ξ , $E|X - \xi|$, is minimized at $\xi = m$, where m is the median of X .*

Proof: We will present the proof in the case of the continuous random variable X with density f . It suffices to show that $E(|X - m|) \leq E(|X - a|)$ for every a . Assume now that $m < a$. Then

$$\begin{aligned} E(|X - m|) - E(|X - a|) &= \int_{-\infty}^m [(m - x) - (a - x)]f(x) dx \\ &\quad + \int_m^a [(x - m) - (a - x)]f(x) dx \\ &\quad + \int_a^{\infty} [(x - m) - (x - a)]f(x) dx, \end{aligned}$$

which simplifies to

$$\int_{-\infty}^m (m - a)f(x) dx + \int_m^a (2x - m - a)f(x) dx + \int_a^{\infty} (a - m)f(x) dx.$$

Since $2x - m - a \leq 2a - m - a = a - m$ for $m \leq x \leq a$, combining the second and third integral, we obtain the inequality

$$E(|X - m|) - E(|X - a|) \leq (m - a)[P(X \leq m) - P(X \geq m)].$$

Since $m - a < 0$ and $P(X \leq m) - P(X \geq m) \geq 1/2 - 1/2 = 0$, we have $E(|X - m|) - E(|X - a|) \leq 0$. The proof for $a < m$ is analogous. $\square$

We now explore the behavior of the variance of sums of random variables. This will naturally lead us to certain new important concepts.

To make the notation more readable, we let $m_X = E(X)$ and $m_Y = E(Y)$. Then

$$\begin{aligned} \text{Var}(X + Y) &= E(X + Y)^2 - [E(X + Y)]^2 \\ &= E(X^2 + 2XY + Y^2) - (m_X + m_Y)^2 \\ &= E(X^2) - m_X^2 + E(Y^2) - m_Y^2 + 2[E(XY) - m_X m_Y] \\ &= \text{Var}(X) + \text{Var}(Y) + 2[E(XY) - m_X m_Y]. \end{aligned}$$

The last quantity appears sufficiently often to deserve a separate name.

Definition 7.6.2 The quantity

$$\text{Cov}(X, Y) = E(XY) - E(X)E(Y) = E[(X - m_X)(Y - m_Y)] \quad (7.34)$$

is called the *covariance* of random variables X and Y . $\square$

We shall show that $\text{Cov}(X, Y)$ exists whenever X and Y have finite second ordinary moments. Indeed, $0 \leq (X - Y)^2 = X^2 - 2XY + Y^2$, and this gives $2XY \leq X^2 + Y^2$. Consequently

$$|XY| \leq \frac{X^2}{2} + \frac{Y^2}{2}.$$

Thus, if $E(X^2) < \infty$ and $E(Y^2) < \infty$, then also $E(|XY|) < \infty$ and $\text{Cov}(X, Y)$ exists.

We have therefore

Theorem 7.6.4 If $E(X_1^2)$ and $E(X_2^2)$ exist, then

$$\text{Var}(X_1 + X_2) = \text{Var}(X_1) + \text{Var}(X_2) + 2\text{Cov}(X_1, X_2),$$

and more generally, if $E(X_i^2) < \infty$ for $i = 1, \dots, n$ then

$$\text{Var}(X_1 + \dots + X_n) = \sum_{j=1}^n \text{Var}(X_j) + 2 \sum_{i < j} \text{Cov}(X_i, X_j).$$

Definition 7.6.3 The random variables X, Y for which $\text{Cov}(X, Y) = 0$ are called uncorrelated or orthogonal. $\square$

Observe that in view of Theorem 7.4.8, independent random variables with finite variances are uncorrelated. Consequently, we have

Theorem 7.6.5 If random variables $X_1, \dots, X_n$ are pairwise uncorrelated, then

$$\text{Var}(X_1 + \dots + X_n) = \text{Var}(X_1) + \dots + \text{Var}(X_n). \quad (7.35)$$

In particular, (7.35) holds if $X_1, \dots, X_n$ are independent.

Let us find the variance of a linear combination of random variables, that is, a variance of the sum

$$Y = a_1 X_1 + a_2 X_2 + \dots + a_n X_n.$$

Using Theorem 7.6.4, we have

$$\text{Var}(Y) = \sum_{i=1}^n \text{Var}(a_i X_i) + 2 \sum_{i < j} \text{Cov}(a_i X_i, a_j X_j).$$

By Theorem 7.6.1, $\text{Var}(a_i X_i) = a_i^2 \text{Var}(X_i)$, while

$$\begin{aligned} \text{Cov}(a_i X_i, a_j X_j) &= E(a_i X_i \times a_j X_j) - E(a_i X_i)E(a_j X_j) \\ &= a_i a_j [E(X_i X_j) - E(X_i)E(X_j)] = a_i a_j \text{Cov}(X_i, X_j). \end{aligned}$$

Consequently,

$$\text{Var}(Y) = \sum_{i=1}^n a_i^2 \text{Var}(X_i) + 2 \sum_{i < j} a_i a_j \text{Cov}(X_i, X_j).$$

In particular, if $X_1, \dots, X_n$ are uncorrelated, then

$$\text{Var}(Y) = \sum_{i=1}^n a_i^2 \text{Var}(X_i). \quad (7.36)$$

■ EXAMPLE 7.37

For the variance of a difference of two random variables, $X_1 - X_2$, we take $a_1 = 1, a_2 = -1$, and obtain

$$\text{Var}(X_1 - X_2) = \text{Var}(X_1) + \text{Var}(X_2) - 2\text{Cov}(X_1, X_2);$$

hence for uncorrelated random variables

$$\text{Var}(X_1 - X_2) = \text{Var}(X_1) + \text{Var}(X_2). \quad (7.37)$$

■ EXAMPLE 7.38 Averaging

If $X_1, \dots, X_n$ are independent, with the same distribution, then in statistics, we call them a *random sample* and their average

$$\bar{X}_n = \frac{X_1 + \dots + X_n}{n}$$

is referred to as a sample mean. If $\text{Var}(X_i) = \sigma^2$, then using $a_i = 1/n, i = 1, \dots, n$ in formula (7.36), we obtain

$$\text{Var}(\bar{X}_n) = \frac{\sigma^2}{n}.$$

Thus, averaging decreases the variability (as measured by the variance) by the factor $1/n$. The standard deviation of the sample mean is therefore decreased (as compared to the standard deviation of a single observation) by the factor $1/\sqrt{n}$.

■ EXAMPLE 7.39 A Foot

The effect of averaging on variability, computed in Example 7.38, appears to have been understood long before the beginning of probability and statistics. This is illustrated by the following law from the Middle Ages that defined the length of one foot (before you read any further, think for a while: How could a measure of length be defined centuries ago so as to be—as much as possible—uniform throughout a country?).

The law specified the following procedure (apparently, the standard of one foot was necessary only on market days—in this case on Sundays after Mass). The shoes of the first 16 men leaving the church (this was an attempt to get a random sample!) lined up, toe to heel (Figure 7.7), gave the “right and lawful rood” (e.g., see Stigler, 1996). Then 1/16 of it was to be used as measure of 1 foot. The number 16 was clearly chosen because it was easy to divide a string into 16 equal parts, by folding it four times into halves.

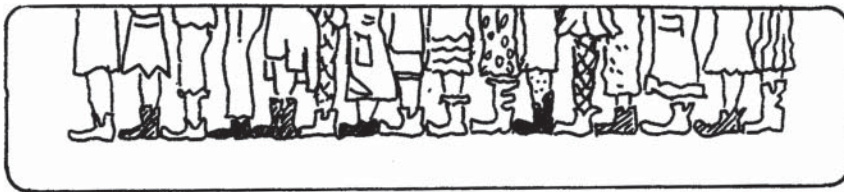


Figure 7.7 Length of 16 feet (Drawing by S. Niewiadomski).

This procedure cuts down the variability of the length of feet, as measured by the standard deviation, by the factor of 4.

■ EXAMPLE 7.40 Problem of Design

Suppose that you have a scale and a set of weights at your disposal. The weight of two objects, A and B, has to be determined as precisely as possible, with the scale used only twice.

SOLUTION. If you put an object on a scale and balance it with weights, you obtain a measurement of the true weight of the object, w , with an error. One of the possible assumptions here, met in many practical situations, is that what one observes is a value of a random variable, X , such that $E(X) = w$ and $\text{Var}(X) = \sigma^2$, with different measurements (even of the same object) being independent. In other words, we have $X = w + \epsilon$, where $E(\epsilon) = 0$, $\text{Var}(\epsilon) = \sigma^2$, with σ being the standard deviation of the error of measurement.

In our situation, it would seem that all one has to do is to put object A on the scale, balance it with weights, and then do the same with object B, observing two random variables, X and Y , with $E(X) = w_A$, $E(Y) = w_B$, and $\text{Var}(X) = \text{Var}(Y) = \sigma^2$.

One can, however, proceed differently. Suppose that on the first weighting, one puts both objects A and B on one scale, and then balances the scales, observing random variable $X_{A+B} = w_A + w_B + \epsilon_1$, where ϵ_1 is the measurement error. On the second weighting, one puts A and B on opposite sides of the scale and adds weights as needed for balance (see Figure 7.8). Thus, on the second measurement, we observe $X_{A-B} = w_A - w_B + \epsilon_2$, where ϵ_2 is independent of ϵ_1 , with $E(\epsilon_1) = E(\epsilon_2) = 0$, $\text{Var}(\epsilon_1) = \text{Var}(\epsilon_2) = \sigma^2$. We easily find that

$$\frac{X_{A+B} + X_{A-B}}{2} = w_A + \frac{\epsilon_1 + \epsilon_2}{2}$$

and

$$\frac{X_{A+B} - X_{A-B}}{2} = w_B + \frac{\epsilon_1 - \epsilon_2}{2}.$$

We have now $\text{Var}[(\epsilon_1 + \epsilon_2)/2] = (1/4)\text{Var}(\epsilon_1 + \epsilon_2) = \sigma^2/2$, and $\text{Var}[(\epsilon_1 - \epsilon_2)/2] = \sigma^2/2$. Using the scale twice, we obtained the measurements of w_A and w_B , each

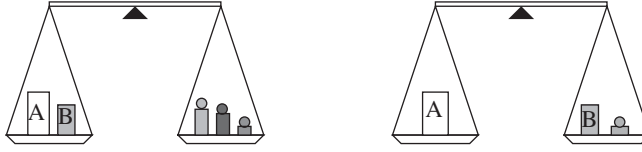


Figure 7.8 Two weightings of A and B.

with standard deviation of the error equal $\sigma/\sqrt{2} = 0.707\sigma$. This means an error reduction by about 30% obtained *at no additional cost* (i.e., with the same number of observations). This is one of the simplest examples of the effects of choosing a proper *design of experiment*.

Let us now investigate the concept of covariance more closely. First, observe that according to formula (7.34),

$$\text{Cov}(X, X) = E[(X - m_X)^2] = \text{Var}(X). \quad (7.38)$$

So, in some sense, covariance is a generalization of variance.

Next, we have the identity describing the behavior of covariance for a linear combination of random variables.

Theorem 7.6.6 *If $X_1, \dots, X_n$ and $Y_1, \dots, Y_m$ are two sets (not necessarily disjoint) of random variables with finite second moments, then for any constants $a_1, \dots, a_n, b_1, \dots, b_m$ and c, d ,*

$$\text{Cov} \left(\sum_{i=1}^n a_i X_i + c, \sum_{j=1}^m b_j Y_j + d \right) = \sum_{i=1}^n \sum_{j=1}^m a_i b_j \text{Cov}(X_i, Y_j). \quad (7.39)$$

The proof involves straightforward checking and will be omitted.

The fact that $\text{Cov}(X, Y) = 0$, if X and Y are independent, suggests using the covariance as a measure of dependence. However, to achieve comparability across various measurements it needs to be standardized. Accordingly, we introduce the following definition:

Definition 7.6.4 Let X and Y be two random variables with finite second moments. Then

$$\rho = \rho_{X,Y} = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)\text{Var}(Y)}}$$

is called the coefficient of *correlation* between X and Y . □

■ EXAMPLE 7.41

Let U, V, W be independent random variables such that $\text{Var}(U) = \sigma_U^2$, $\text{Var}(V) = \sigma_V^2$, and $\text{Var}(W) = \sigma_W^2$. Compute the coefficient of correlation between $X = U + W$ and $Y = V + W$.

SOLUTION. We have $\sigma_X^2 = \sigma_U^2 + \sigma_W^2$ and $\sigma_Y^2 = \sigma_V^2 + \sigma_W^2$, whereby $\text{Cov}(X, Y) = \text{Cov}(U + W, V + W) = \text{Cov}(U, V) + \text{Cov}(U, W) + \text{Cov}(W, V) + \text{Cov}(W, W) = \sigma_W^2$ by (7.38). Consequently,

$$\rho_{XY} = \frac{\sigma_W^2}{\sqrt{(\sigma_U^2 + \sigma_W^2)(\sigma_V^2 + \sigma_W^2)}}. \quad (7.40)$$

Situations like this are quite common; variables often are related because they are influenced by a common other variable (W). One can think about the prices of two items X and Y that depend on the price W of some components, which are used in both X and Y . To see the effect of σ_W^2 , let us write (7.40) as

$$\rho_{XY} = \frac{1}{\sqrt{\left(\frac{\sigma_U^2}{\sigma_W^2} + 1\right) \left(\frac{\sigma_V^2}{\sigma_W^2} + 1\right)}}.$$

If σ_W^2 is large compared with both σ_U^2 and σ_V^2 , then $\rho_{X,Y}$ is close to 1. On the other hand, if σ_W^2 is small compared to one (or both) of σ_U^2 and σ_V^2 , then $\rho_{X,Y}$ is close to 0. These results are expected of $\rho_{X,Y}$ as a measure of dependence of random variables X and Y .

■ EXAMPLE 7.42

Let X, Y have a joint density uniform on the square with vertices $(-1, 0)$, $(0, 1)$, $(1, 0)$, and $(0, -1)$ (see Figure 7.9).

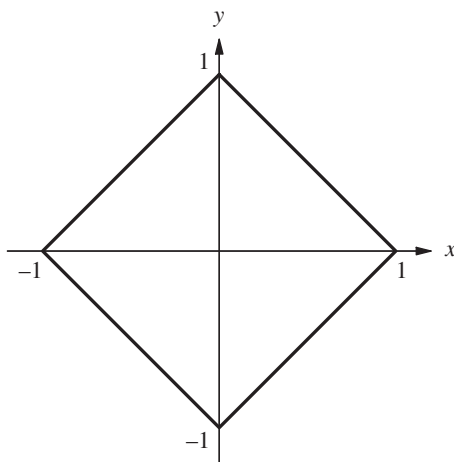


Figure 7.9 Dependent but uncorrelated random variables.

Clearly, $E(X) = E(Y) = 0$ by the symmetry of marginal distributions. Also $E(XY) = 0$, since in the integral $\int \int xyf(x, y) dx dy$ the contribution arising from domains with $xy > 0$ and $xy < 0$ cancel one another. Thus, $\text{Cov}(X, Y) = 0$, and therefore $\rho_{X,Y} = 0$. Yet the random variables are dependent. This can be seen immediately from Theorem 6.2.2, as well as from the fact that if one knows, for example, that X is close to 1, then one can infer that Y must be close to zero.

This example shows that correlation, as a measure of dependence, is not perfect. There are dependent random variables for which $\rho_{X,Y} = 0$. To analyze the properties of the correlation coefficient, we need to introduce the following inequality:

Theorem 7.6.7 (Schwarz Inequality) For any variables X and Y ,

$$[E(XY)]^2 \leq E(X^2)E(Y^2). \quad (7.41)$$

Proof: If $E(X^2)$ or $E(Y^2)$ is infinite, the inequality (7.41) holds. On the other hand, if $E(X^2) = 0$, then $P(X = 0) = 1$, $P(XY = 0) = 1$, and $E(XY) = 0$. Hence, again, inequality (7.41) holds. The same argument applies if $E(Y^2) = 0$. So we can assume that $0 < E(X^2) < \infty$ and $0 < E(Y^2) < \infty$.

Consider now the random variable $Z_t = tX + Y$. For every t , we have

$$\begin{aligned} 0 \leq E(Z_t^2) &= E(t^2X^2 + 2tXY + Y^2) \\ &= t^2E(X^2) + 2tE(XY) + E(Y^2). \end{aligned} \quad (7.42)$$

The right-hand side is a quadratic function of t , which is nonnegative for all t . Thus, its discriminant must satisfy the condition

$$4[E(XY)]^2 - 4E(X^2)E(Y^2) \leq 0$$

which is the same as (7.41). $\square$

We can now prove the following theorem, asserting the basic properties of the correlation coefficient:

Theorem 7.6.8 *The coefficient of correlation ρ_{XY} satisfies the inequality*

$$-1 \leq \rho_{X,Y} \leq 1 \quad (7.43)$$

with $|\rho_{X,Y}| = 1$ if, and only if, there exist constants a ($a \neq 0$) and b such that $P\{Y = aX + b\} = 1$.

Moreover, $|\rho|$ is invariant under linear transformation of random variables; more precisely, for any a, b, c, d with $ac \neq 0$,

$$\rho_{aX+b, cY+d} = \epsilon \rho_{X,Y} \quad (7.44)$$

with $\epsilon = +1$ if $ac > 0$ and $\epsilon = -1$ if $ac < 0$.

Proof: By formula (7.39), we have $\text{Cov}(aX + b, cY + d) = ac\text{Cov}(X, Y)$. On the other hand, $\text{Var}(aX + b) = a^2\text{Var}(X)$, $\text{Var}(cY + d) = c^2\text{Var}(Y)$, so that

$$\rho_{aX+b, cY+d} = \frac{ac\text{Cov}(X, Y)}{\sqrt{a^2\text{Var}(X) \times c^2\text{Var}(Y)}} = \frac{ac}{|ac|} \times \rho_{X,Y},$$

which proves (7.44). We can now prove (7.43). By what is already shown in (7.39), we can assume that $EX = EY = 0$, so

$$\rho_{X,Y} = E(XY) / \sqrt{E(X^2)E(Y^2)}.$$

The condition $|\rho_{X,Y}| \leq 1$ is equivalent to the Schwarz inequality (7.41).

It remains to prove that $\rho^2 = 1$ is equivalent to the existence of a linear relationship between X and Y . Again, we can assume that $E(X) = E(Y) = 0$. If $Y = aX$ with $a \neq 0$, then $E(XY) = E[X(aX)] = aE(X^2)$ and $E(X)E(Y) = a[E(X)]^2$. Consequently, $\text{Cov}(X, Y) = a\{E(X^2) - [E(X)]^2\} = a\text{Var}(X)$, and since $\text{Var}(Y) = a^2\text{Var}(X)$, we get

$$\rho_{X,Y} = \frac{a\text{Var}(X)}{\sqrt{\text{Var}(X) \times a^2\text{Var}(X)}} = \frac{a}{|a|} = \pm 1.$$

Conversely, assume that $\rho_{X,Y}^2 = 1$. Then (assuming $E(X) = E(Y) = 0$) we have $[E(XY)]^2 = E(X^2)E(Y^2)$, meaning X and Y such that we have equality in (7.42). The proof of Theorem 7.6.7 shows that this occurs if the discriminant of the

right-hand side of (7.42) is 0, hence if there exists t^* such that $E(t^*X + Y)^2 = 0$. But the expectation of a nonnegative random variable is 0 if, and only if, this variable assumes only the value zero, so that $P(t^*X + Y = 0) = 1$. If we had $t^* = 0$, then we would have $P(Y = 0) = 1$, and hence $E(Y^2) = 0$, a case that we eliminated. This shows that $t^* \neq 0$ and that there is a linear relationship between X and Y . $\square$

■ EXAMPLE 7.43

Consider the situation where the value of some random variable Z has to be predicted based on the values of two other variables X and Y .

The optimal solution to this problem is well known. Adapting Theorem 7.6.2 to the case of conditional prediction, we see that given $X = x$ and $Y = y$, one should predict the conditional expectation $\xi = E(Z|X = x, Y = y)$. However, the practical implementation of this solution presupposes the knowledge of the joint distribution of (X, Y, Z) , and—which may be analytically difficult—the conditional expectation of Z given (X, Y) .

Quite often we simply do not have this information. If appropriate, one can then use here the best *linear* prediction, which requires the knowledge of the first two moments of the joint distribution of (X, Y, Z) only, that is, expectations, variances, and covariances (or equivalently, correlations).

SOLUTION. Without loss of generality, we can assume that $E(X) = E(Y) = E(Z) = 0$ (since if we know the mean, predicting the random variable is equivalent to predicting its deviation from the mean). We will be looking for a predictor of the form $\xi = \alpha X + \beta Y$ that minimizes the expected square error

$$E(Z - \xi)^2 = E(Z - \alpha X - \beta Y)^2. \quad (7.45)$$

Condition (7.45) can be expanded to

$$\begin{aligned} E(Z - \alpha X - \beta Y)^2 &= E(Z^2) + \alpha^2 E(X^2) + \beta^2 E(Y^2) \\ &\quad - 2\alpha E(XZ) - 2\beta E(YZ) + 2\alpha\beta E(XY) \\ &= \sigma_Z^2 + \alpha^2 \sigma_X^2 + \beta^2 \sigma_Y^2 \\ &\quad - 2\alpha \rho_{XZ} \sigma_Z \sigma_X - 2\beta \rho_{YZ} \sigma_Z \sigma_Y + 2\alpha\beta \rho_{XY} \sigma_X \sigma_Y. \end{aligned}$$

Differentiating with respect to α and β and setting the derivatives equal to zero, we obtain the system of linear equations:

$$\begin{aligned} \alpha \sigma_X + \beta \rho_{XY} \sigma_Y &= \rho_{XZ} \sigma_Z, \\ \alpha \rho_{XY} \sigma_X + \beta \sigma_Y &= \rho_{YZ} \sigma_Z. \end{aligned}$$

The solution exists if $\rho_{XY}^2 \neq 1$:

$$\alpha = \frac{\rho_{XZ} - \rho_{XY} \rho_{YZ}}{1 - \rho_{XY}^2} \times \frac{\sigma_Z}{\sigma_X} \quad \text{and} \quad \beta = \frac{\rho_{YZ} - \rho_{XY} \rho_{XZ}}{1 - \rho_{XY}^2} \times \frac{\sigma_Z}{\sigma_Y}$$

(if $\rho_{XY}^2 = 1$, then $X = Y$ or $X = -Y$, so X and Y provide the same information). This solution is not very important by itself. More important is the fact that this method can easily be applied to determine the best linear predictor of one or more random variables. All that we need are the means and second order moments of all variables in question—those to be predicted and those serving as predictors. The next example

will illustrate the class of situations in which such a prediction method can be usefully applied.

■ EXAMPLE 7.44 Moving Averages

Consider the process $\{X_t\}$ of prices of some commodity. In many cases, the process $\{X_t\}$ is subject to seasonal variations. For example, if t is measured in months, and X_t represents the (average) monthly price of tomatoes, then X_t varies with the period of 12. One of the methods of detecting trends (or prediction), which takes such periodic seasonal variation into account, is based on taking the averages of X_t over the most recent complete period. Thus, we can define

$$Y_t = \frac{X_t + X_{t-1} + \cdots + X_{t-11}}{12}.$$

Here a possible assumption about X_t may be $X_t = f_t + \xi_t$, where f_t is some non-random function with period 12 (i.e., $f_t = f_{t-12}$ for every t) and ξ_t s are independent random variables with the same distribution (sometimes referred to as “shocks”) representing random effects. Under these assumptions,

$$Y_t = C + \frac{\xi_t + \xi_{t-1} + \cdots + \xi_{t-11}}{12},$$

where $C = \frac{1}{12}(f_t + f_{t-1} + \cdots + f_{t-11})$, which is a constant independent of t . Clearly, we can assume, without loss of generality, that $E(\xi_t) = 0$ for all t . Let $\text{Var}(\xi_t) = E(\xi_t^2) = \sigma^2 > 0$. Then $E(Y_t) = C$, $\text{Var}(Y_t) = \sigma^2/12$ and

$$\begin{aligned} \text{Cov}(Y_t, Y_{t-m}) &= \text{Cov}\left(\frac{\xi_t + \xi_{t-1} + \cdots + \xi_{t-11}}{12}, \frac{\xi_{t-m} + \cdots + \xi_{t-m-11}}{12}\right) \\ &= \frac{1}{144} \text{Cov}(\xi_t + \cdots + \xi_{t-11}, \xi_{t-m} + \cdots + \xi_{t-m-11}). \\ &= K \times \frac{\sigma^2}{144}, \end{aligned}$$

where K is the number of overlapping terms in sums $\xi_t + \xi_{t-1} + \cdots + \xi_{t-11}$ and $\xi_{t-m} + \cdots + \xi_{t-m-11}$. Obviously, the number of such terms is 0 if $m \geq 12$ and equals $12 - m$ otherwise.

Consequently, we have

$$\text{Cov}(Y_t, Y_{t-m}) = \max\left(0, \frac{12-m}{144} \sigma^2\right)$$

so that

$$\rho_{Y_t, Y_{t-m}} = \begin{cases} 1 - \frac{m}{12} & \text{for } m = 0, 1, \dots, 12 \\ 0 & \text{for } m \geq 12. \end{cases} \quad (7.46)$$

PROBLEMS

- 7.6.1** A random variable X has binomial distribution with mean 5 and standard deviation 2. Find $P\{X = 6\}$.
- 7.6.2** Find the variance of variable X if its first ordinary moment is 3 and the second factorial moment is 52.

7.6.3 For random variables X and Y such that $E(X) = 2, E(Y) = 1, E(X^2) = 10, E(Y^2) = 3$, and $E(XY) = c$, find: **(i)** $\text{Var}(3X - 5Y)$. **(ii)** $\rho_{X,Y}$. **(iii)** The range of values of c for which the assumptions of the problem are consistent.

7.6.4 Find the variance of a random variable X with a cdf

$$F(x) = \begin{cases} 0 & \text{for } x < 0 \\ \sqrt{x} & \text{for } 0 \leq x \leq 1 \\ 1 & \text{for } x > 1. \end{cases}$$

7.6.5 Let $m_X(t) = p + 0.5e^{-t} + (0.5 - p)e^t$ be a mgf of a variable X . Determine possible values of p and find $\text{Var}(X^{27})$ as a function of p .

7.6.6 Random variables X and Y are jointly distributed with the density $f(x, y) = (24/11)y(1 - x - y)$ for $x > 0, y > 0$, and $x + y < 1$. Find the $\text{Cov}(X, Y)$.

7.6.7 Let variables X and Y be such that $E(X) = E(Y) = 0, \text{Var}(X) = \text{Var}(Y) = 1$, and $\rho_{X,Y} = \rho$. Find $E(W), \text{Var}(W)$, and $\rho_{W,Y}$ if $W = X - \rho Y$.

7.6.8 Find the correlation of random variables X and Y jointly uniformly distributed on: **(i)** The triangle with vertices $(0, 0), (1, 0), (1, 2)$. **(ii)** The quadrangle with vertices $(0, 0), (a, 0), (a, 2)$, and $(2a, 2)$, where $a > 0$.

7.6.9 Let X, Y be independent, with means μ_X, μ_Y and variances σ_X^2, σ_Y^2 . Show that $\text{Var}(XY) = \sigma_X^2 \sigma_Y^2 + \sigma_X^2 \mu_Y^2 + \sigma_Y^2 \mu_X^2$.

7.6.10 Let $X_1, \dots, X_n$ be independent random variables having the same distribution with a mean μ and variance σ^2 . Let $\bar{X} = (X_1 + \dots + X_n)/n$. Show that $E\{\sum_{i=1}^n (X_i - \bar{X})^2\} = (n-1)\sigma^2$. [Hint: Since $X_i - \bar{X} = (X_i - \mu) - (\bar{X} - \mu)$, we have $\sum_{i=1}^n (X_i - \bar{X})^2 = \sum_{i=1}^n (X_i - \mu)^2 - n(\bar{X} - \mu)^2$.]

7.7 CONDITIONAL EXPECTATION

The concept of conditional probability was introduced in Chapter 4 and then extended to the concept of conditional distribution of a random variable in Chapters 5 and 6. Now, we will introduce the expectation of this distribution as a number, $E(X|Y = y)$, determined for every particular value $Y = y$, provided that it exists.

Definition 7.7.1 For two random variables X and Y , the *conditional expectation* $E(X|Y)$ is defined as the random variable that assumes the value $E(X|Y = y)$ when $Y = y$. $\square$

■ EXAMPLE 7.45

Let Y have $\text{POI}(\lambda)$ distribution, and given $Y = n$, let X have $\text{BIN}(n, p)$ distribution. We can think here (recall Example 6.14) of Y being the number of eggs laid by a bird and of X as the number of those eggs that hatch (assuming that the eggs hatch independently, each with probability p). This means that the expected number of eggs that hatch, if $Y = n$, is $E(X|Y = n) = np$.

■ EXAMPLE 7.46

Let (X, Y) have a distribution uniform on the triangle with vertices $(0, 0), (0, 1)$, and $(1, 1)$. Given Y , the distribution of X is uniform on the intersection of the line $Y = y$

and the support of the density. Since the expectation of the uniform distribution on an interval is its midpoint, we have $E(X|Y = y) = y/2$, and similarly $E(Y|X = x) = (1 + x)/2$, where $0 \leq X \leq 1, 0 \leq Y \leq 1$.

■ **EXAMPLE 7.47**

Consider now a case where one variable is discrete and the other is continuous. Let $P\{Y = 1\} = \alpha$ and $P\{Y = 2\} = 1 - \alpha$, and given Y , let X have the density

$$f(x|Y = y) = yx^{y-1}, \quad 0 \leq x \leq 1.$$

In this case, if $Y = 1$, then X is uniform on $[0, 1]$; hence $E(X|Y = 1) = 1/2$. If $Y = 2$, then X has density $f(x|Y = 2) = 2x$ on $[0, 1]$; hence $E(X|Y = 2) = 2/3$. On the other hand, given $X = x$, we have

$$P\{Y = 1|X = x\} = \frac{\alpha}{\alpha + 2x(1 - \alpha)},$$

$$P\{Y = 2|X = x\} = \frac{2x(1 - \alpha)}{\alpha + 2x(1 - \alpha)}.$$

This is a consequence of Bayes' formula, interpreting the values of the density $f(x|Y)$ as infinitesimal probability $P\{X = x|Y\} dx$. Thus

$$\begin{aligned} E(Y|X = x) &= 1 \times P(Y = 1|X = x) + 2 \times P(Y = 2|X = x) \\ &= \frac{\alpha + 4x(1 - \alpha)}{\alpha + 2x(1 - \alpha)}. \end{aligned}$$

We will now prove

Theorem 7.7.1 *For any random variables X, Y , we have*

$$E[E(X|Y)] = E(X), \tag{7.47}$$

provided that $E(X)$ exists.

Proof: We will present the proof only in the case of continuous random variables. Letting $f_1(x)$ and $f_2(y)$ denote the marginal densities of the joint density $f(x, y)$, we have

$$E(X|Y = y) = \int xg(x|Y = y) dx = \int x \frac{f(x, y)}{f_2(y)} dx.$$

Hence, taking expectation with respect to $Y = y$, we obtain

$$\begin{aligned} E[E(X|Y)] &= \int \left[\int \frac{xf(x, y)}{f_2(y)} dx \right] f_2(y) dy = \int x \left[\int f(x, y) dy \right] dx \\ &= \int xf_1(x) dx = E(X). \end{aligned}$$

The interchange of the order of integration is legitimate in view of the assumption that $E(X)$ exists. □

■ **EXAMPLE 7.48**

In Example 7.45 we had $E(X|Y) = Yp$. Since Y has Poisson distribution with mean λ , we obtain $E(X) = E(Y) \times p = \lambda p$. This result also follows from the fact (see Example 6.14) that X has Poisson distribution with mean λp .

■ **EXAMPLE 7.49**

In Example 7.46, we have $E(X) = E[E(X|Y)] = E(Y/2)$ and $E(Y) = E[E(Y|X)] = E[(1+X)/2]$, which gives $E(X) = E(Y)/2$, $E(Y) = 1/2 + E(X)/2$. This system of two equations can easily be solved, leading to $\mu_X = 1/3$ and $\mu_Y = 2/3$.

■ **EXAMPLE 7.50**

Finally, in Example 7.47, we have

$$E(X) = E[E(X|Y)] = \alpha E(X|Y=1) + (1-\alpha)E(X|Y=2) = \frac{1}{2}\alpha + \frac{2}{3}(1-\alpha).$$

On the other hand, since $f_1(x) = \alpha + 2x(1-\alpha)$, we have

$$\begin{aligned} E(Y) &= E[E(Y|X)] = \int_0^1 \frac{\alpha + 4x(1-\alpha)}{\alpha + 2x(1-\alpha)} f_1(x) dx \\ &= \int_0^1 [\alpha + 4x(1-\alpha)] dx = 2 - \alpha. \end{aligned}$$

Theorem 7.7.1 can be very helpful in determining the expectation of a random variable X by conditioning it with respect to some other variable (Y). The choice of Y is crucial here if a simplification is to be achieved.

The following theorem is an analogue of Theorem 7.7.1:

Theorem 7.7.2 *For any random variables X, Y , if $E(X^2)$ exists, then*

$$\text{Var}(X) = E[\text{Var}(X|Y)] + \text{Var}[E(X|Y)]. \quad (7.48)$$

Proof: Let $\mu_X = E(X)$. We have

$$\begin{aligned} \text{Var}(X) &= E(X - \mu_X)^2 = E\{[X - E(X|Y)] + [E(X|Y) - \mu_X]\}^2 \\ &= E\{[X - E(X|Y)]^2\} + 2E\{[X - E(X|Y)] \times [E(X|Y) - \mu_X]\} \\ &\quad + E\{[E(X|Y) - \mu_X]^2\} \\ &= A + B + C. \end{aligned}$$

Using Theorem 7.7.1, we obtain

$$\begin{aligned} A &= E\{E[X - E(X|Y)]^2|Y\} = E[\text{Var}(X|Y)], \\ C &= E[E(X|Y) - \mu_X]^2 = \text{Var}[E(X|Y)], \end{aligned}$$

where the last equality follows from the fact that $\mu_X = E[E(X|Y)]$ (see (7.47)). It remains to prove that $B = 0$.

As before, we have

$$\begin{aligned}
 B &= E\{[X - E(X|Y)][E(X|Y) - \mu_X]\} \\
 &= E(E\{[X - E(X|Y)] \times [E(X|Y) - \mu_X] | Y\}) \\
 &= E(E[U(X, Y) \times V(Y) | Y]),
 \end{aligned}$$

where $U(X, Y) = X - E(X|Y)$ and $V(Y) = E(X|Y) - \mu_X$ is constant for every fixed value of Y . Consequently, $V(Y)$ can be factored out, leaving

$$B = E\{V(Y)E[U(X, Y)|Y]\}.$$

But $E[U(X, Y)|Y] = E[X - E(X|Y)|Y] = E(X|Y) - E(X|Y) = 0$ and therefore $B = 0$, which completes the proof. $\square$

■ EXAMPLE 7.51

For an application of Theorem 7.7.2, let us return again to the Example 7.45. We have there $X \sim \text{BIN}(Y, p)$, so $E(X|Y) = Yp$, and $\text{Var}(X|Y) = Ypq$. Consequently, since Y has $\text{POI}(\lambda)$ distribution with $E(Y) = \text{Var}(Y) = \lambda$, we obtain

$$\begin{aligned}
 \text{Var}(X) &= E\text{Var}(X|Y) + \text{Var}[E(X|Y)] = E(Ypq) + \text{Var}(Yp) \\
 &= pq\lambda + p^2\lambda = \lambda p,
 \end{aligned}$$

again, in agreement with our finding from Example 6.14 that the marginal distribution of X is Poisson with a parameter λp .

Finally, as another example of an application, we will prove the theorem due to Hotelling and Solomons (1932) that connects the mean, median, and standard deviation of any random variable. The present proof is a slight modification of the proof by O'Cinneide (1990).

Theorem 7.7.3 *Let X be a random variable with $E(X^2) < \infty$, and let μ and σ be its mean and standard deviation, respectively. Moreover, let m be any median of X , that is, a number such that $P(X \geq m) \geq 0.5$ and $P(X \leq m) \geq 0.5$. Then for any random variable X ,*

$$|\mu - m| \leq \sigma;$$

that is, the mean is within one standard deviation of any median.

Proof: Let $\pi^- = P(X < m)$, $\pi^+ = P(X > m)$, and $\pi^0 = P(X = m)$ so that $\pi^- + \pi^0 \geq 0.5$ and $\pi^0 + \pi^+ \geq 0.5$. Let Y be the random variable defined as follows:

(a) If $\pi^0 = 0$, then

$$Y = \begin{cases} 1 & \text{if } X < m \\ 2 & \text{if } X > m. \end{cases} \quad (7.49)$$

(b) If $\pi^0 > 0$, then in addition to (7.49), we let

$$P\{Y = 1|X = m\} = 0.5 - \pi^- \quad \text{and} \quad P\{Y = 2|X = m\} = 0.5 - \pi^+.$$

Clearly, $P\{Y = 1\} = P\{Y = 2\} = 0.5$.

For simplicity, put $\mu_i = E(X|Y = i)$, for $i = 1, 2$. Then $\mu = E(X) = E[E(X|Y)] = 0.5E(X|Y = 1) + 0.5E(X|Y = 2)$, that is,

$$\mu = 0.5\mu_1 + 0.5\mu_2. \quad (7.50)$$

Assume that $m \leq \mu$ (the argument in the case of opposite inequality is analogous). Clearly, we have $\mu_1 \leq m$, which in view of (7.50) gives the inequality

$$\mu - m \leq \mu - \mu_1 = 0.5(\mu_2 + \mu_1) - \mu_1 = 0.5(\mu_2 - \mu_1). \quad (7.51)$$

Using Theorem 7.7.2 we can now write

$$\begin{aligned} \sigma^2 &= \text{Var}(X) = E[\text{Var}(X|Y)] + \text{Var}[E(X|Y)] \geq \text{Var}[E(X|Y)] \\ &= \left(\frac{\mu_1 - \mu_2}{2} \right)^2 \geq (\mu - m)^2, \end{aligned}$$

and $\sigma \geq |\mu - m|$, as asserted. $\square$

PROBLEMS

7.7.1 Variables X and Y are jointly distributed with the density $f(x, y) = Cx(3x + 2y)$ for $0 < x < y < 1$, and 0 otherwise. Find: **(i)** C . **(ii)** $P(X > 0.5)$. **(iii)** $E(Y|X = 0.3)$.

7.7.2 Let X and Y have the joint density uniform on the triangle with vertices $(0, 0)$, $(2, 0)$ and $(3, 1)$. Find: **(i)** $E(X|Y)$ and $E(Y|X)$. **(ii)** $\text{Var}(X|Y)$ and $\text{Var}(Y|X)$. **(iii)** The expectations and variances of X and Y using formulas (7.47) and (7.48).

7.7.3 Let X, Y be continuous random variables with a joint density $f(x, y)$. Assume that $E(Y|X = x) = \mu$ for all x . Show that

$$\text{Var}(Y) = \int \text{Var}(Y|X = x) f_X(x) dx.$$

7.7.4 The number of traffic accidents that occur in a certain city in a week is a random variable with mean μ and variance σ^2 . The numbers of people injured in an accident are independent random variables, each with mean m and variance k^2 . Find the mean and variance of the number of people injured in a week.

7.7.5 Let $X_1, \dots, X_n$ be independent variables with a $U[0, 1]$ distribution. The joint density of the $S = \min(X_1, \dots, X_n)$ and $T = \max(X_1, \dots, X_n)$ is $f(s, t) = n(n-1)(t-s)^{n-2}$ for $0 \leq s \leq t \leq 1$ and $f(s, t) = 0$ otherwise. Find: **(i)** $E(S^m)$, $E(T^m)$, and $\rho(S, T)$. **(ii)** $E(T|S)$.

7.8 INEQUALITIES

In this section, we will introduce three important inequalities, all involving expectations, that will be used in the following chapters.

Theorem 7.8.1 If V is a random variable such that $E(V) = 0$ and $\text{Var}(V) = 1$, then for every $t > 0$,

$$P\{|V| \geq t\} \leq 1/t^2. \quad (7.52)$$

Proof: We will give the proof in the case of continuous random variables; the proof in the discrete case (as well as in the general case) will require only notational changes. Letting f denote the density of V , we have

$$\begin{aligned} 1 = \text{Var}(V) &= E(V^2) = \int_{-\infty}^{+\infty} v^2 f(v) dv \geq \int_{|v| \geq t} v^2 f(v) dv \\ &\geq t^2 \int_{|v| \geq t} f(v) dv = t^2 P\{|V| \geq t\}, \end{aligned}$$

which was to be shown. $\square$

Clearly, inequality (7.52) is not informative for $t \leq 1$.

If X is a random variable with finite positive variance $\text{Var}(X) = \sigma^2 > 0$ and $\mu = E(X)$, we can always standardize X by defining

$$V = \frac{X - \mu}{\sigma}. \quad (7.53)$$

Obviously, $E(V) = 0$ and $\text{Var}(V) = 1$; transformation of X into V amounts to introducing a new scale of expressing values of X , with the origin of the scale located at $E(X)$, and the unit of measurement being the standard deviation σ . Applying formula (7.52) to random variable (7.53), we obtain the Chebyshev inequality:

Theorem 7.8.2 (Chebyshev Inequality) *If $\text{Var}(X) = \sigma^2 < \infty$, then for every $t > 0$,*

$$P\{|X - \mu| \geq t\sigma\} \leq \frac{1}{t^2} \quad (7.54)$$

or for $\epsilon = t\sigma$,

$$P\{|X - \mu| \geq \epsilon\} \leq \frac{\sigma^2}{\epsilon^2}. \quad (7.55)$$

Both (7.55) and (7.54) show the role of the standard deviation and variance. We obtain the bounds on probabilities of random variable X deviating from its mean μ by more than a certain amount, which can be expressed as a multiple of σ in (7.54) or in original units in (7.55).

■ EXAMPLE 7.52

For any random variable, the probability that it will deviate from its mean by more than three standard deviations is, at most, $1/9$. This probability, however, can be much less than $1/9$ for some random variables. For instance, it equals 0.0026 for a normal distribution; see (8.60).

Consequently, the *three-sigma rule* (which says that we can practically disregard the probability that a random variable will deviate from its mean by more than three standard deviations) should be used with caution. It may safely be used for random variables with either a normal distribution, or close to normal, but in the general case, the probability $1/9$ can hardly be disregarded.

We will now find the bounds given by the Chebyshev inequality in the case of few selected distributions.

■ **EXAMPLE 7.53**

Suppose that we toss a coin 20 times. Let us estimate the probability that the number of heads will deviate from 10 by 3 or more (then we will have at most 7 or at least 13 heads). Here $\mu = np = 10$, $\sigma^2 = npq = 5$ (see Examples 7.3 and 7.33). We have

$$P\{|X - 10| \geq 3\} \leq \frac{\sigma^2}{3^2} = \frac{5}{9} = 0.555.$$

The exact probability equals 0.2632.

■ **EXAMPLE 7.54**

Let X have exponential distribution with density $f(x) = \lambda e^{-\lambda x}$, $x > 0$. We have here $\mu = \sigma = 1/\lambda$ (see Example 7.35). Consequently,

$$P\left\{\left|X - \frac{1}{\lambda}\right| > \frac{t}{\lambda}\right\} \leq \frac{1}{t^2}.$$

For instance, if $t = 1$ we obtain a noninformative bound 1, while the probability that X will deviate from $1/\lambda$ by more than $1/\lambda$ is (remember that exponential random variable can assume only positive values)

$$P\left\{X > \frac{2}{\lambda}\right\} = e^{-\lambda(2/\lambda)} = e^{-2} = 0.13534.$$

As can be seen from these examples, the quality of bounds given by the Chebyshev inequality is not impressive. However, the most important thing here is that the Chebyshev inequality gives a universal bound, valid for *all* random variables with finite variance. In fact the bound as such cannot be improved.

Among the most important consequences of the Chebyshev inequality are the so-called laws of large numbers. We explore this topic in some detail in Chapter 9. Here we illustrate the situation by the following example:

■ **EXAMPLE 7.55 Binomial Distribution**

Consider the binomial random variable S_n = number of successes in n trials. We have then $E(S_n) = np$ and $\text{Var}(S_n) = np(1-p)$ when p is the probability of success. Consequently, $E(S_n/n) = p$, $\text{Var}(S_n/n) = pq/n$, and the Chebyshev inequality gives

$$P\left\{\left|\frac{S_n}{n} - p\right| \geq \epsilon\right\} \leq \frac{pq}{n\epsilon^2}.$$

Letting $n \rightarrow \infty$ we obtain the following theorem:

Theorem 7.8.3 *If S_n has binomial distribution with parameter p , then for every $\epsilon > 0$*

$$\lim_{n \rightarrow \infty} P\left\{\left|\frac{S_n}{n} - p\right| \geq \epsilon\right\} = 0.$$

This theorem appears to explain why the empirical frequency of an event, namely S_n/n , approaches, as the number of trials n increases, the probability p of the event. It tells us that

for any positive number ϵ it becomes increasingly unlikely that the empirical frequency will deviate from theoretical probability by more than ϵ . The Chebyshev inequality assumes that the random variable has finite second moment. One occasionally needs a bound for a tail of the distribution without this assumption. In such cases, we have the following theorem:

Theorem 7.8.4 (Markov Inequality) *If X is a nonnegative random variable with $E(X) < \infty$, then for every $t > 0$,*

$$P\{X > t\} \leq \frac{E(X)}{t}. \quad (7.56)$$

Proof: We give the proof for the discrete case. Let $x_1, x_2, \dots$ be possible values of X . Then

$$\begin{aligned} E(X) &= \sum_i x_i P(X = x_i) \geq \sum_{x_i > t} x_i P(X = x_i) \\ &\geq t \sum_{x_i > t} P(X = x_i) = tP(X > t). \end{aligned}$$

□

The last inequality will be given without proof:

Theorem 7.8.5 (Kolmogorov Inequality) *Let independent random variables $X_1, X_2, \dots$ have $E(X_i) = 0$ and finite variances $\sigma_j^2, j = 1, 2, \dots$ (in the case of nonzero means we can always consider new variables $X'_j = X_j - E(X_j)$ with means that equal zero). If $S_j = X_1 + X_2 + \dots + X_j$, then for every $t > 0$,*

$$P\{\max_{1 \leq j \leq n} |S_j| \geq t\} \leq \frac{\text{Var}(S_n)}{t^2}. \quad (7.57)$$

PROBLEMS

7.8.1 Let X be any random variable. Show that $E(X) \geq 0.2$ if $P(X \geq 0) = 1$ and $P(X \geq 2) = 0.1$.

7.8.2 Assume that $E(X) = 12$, $P(X \geq 14) = 0.12$, and $P(X \leq 10) = 0.18$. Show that $\text{Var}(X)$ is at least 1.2.

7.8.3 Prove the Markov inequality when X is a continuous random variable.

7.8.4 Derive the Chebyshev inequality from the Markov inequality.

7.8.5 Show that if X has a mgf bounded by the mgf of exponential distribution (i.e., $\lambda/(\lambda - t)$ for $t < \lambda$), then for $\lambda \epsilon > 1$ we have

$$P\{X > \epsilon\} \leq \lambda \epsilon e^{-(\lambda \epsilon - 1)}.$$

(Hint: Use the mgf of exponential distribution to obtain the bound for $P\{X > \epsilon\}$, then determine its minimum.)

7.8.6 Let X have the Poisson distribution with mean λ . Show that

$$P\left\{X \leq \frac{\lambda}{2}\right\} \leq \frac{4}{\lambda} \quad \text{and} \quad P\{X \geq 2\lambda\} \leq \frac{1}{\lambda}.$$

- 7.8.7** Let X be a random variable such that a mgf $m_X(t)$ exists for all t . Use the same argument as in the proof of the Chebyshev inequality to show that $P\{X \geq y\} \leq e^{-ty}m_X(t), t \geq 0$.
- 7.8.8** Show that if X has the Poisson distribution with mean λ , then $P(X \leq \lambda/2) \leq (2/e)^{\lambda/2}$ and $P(X \geq 2\lambda) \leq (e/4)^\lambda$. (*Hint*: Use the inequality in Problem 7.8.7, and find minimum of the right-hand sides for t .)